Integrative Approach to Assess Anti-Inflammatory Effects of Combination of Curcumin and Nimbidin in Experimental Animals

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In the subject

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DECLARATION BY THE CANDIDATE

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List of Abbreviations

AA – Arachidonic Acid

AKT – Protein Kinase B

CAT – Catalase

COX – Cyclooxygenase

COX-2 – Cyclooxygenase-2

DNA – Deoxyribonucleic Acid

ERK – Extracellular Signal-Regulated Kinase

GSH – Reduced Glutathione

IL – Interleukin

IL-1β – Interleukin-1 beta

IL-6 – Interleukin-6

iNOS – Inducible Nitric Oxide Synthase

JAK – Janus Kinase

JNK – c-Jun N-terminal Kinase

LOX – Lipoxygenase

LT – Leukotriene

MAPK – Mitogen-Activated Protein Kinase

MDA – Malondialdehyde

NF-κB – Nuclear Factor kappa B

NO – Nitric Oxide

PGE2 – Prostaglandin E2

PI3K – Phosphoinositide 3-Kinase

PPI – Protein–Protein Interaction

ROS – Reactive Oxygen Species

SOD – Superoxide Dismutase

STAT – Signal Transducer and Activator of Transcription

TNF-α – Tumor Necrosis Factor-alpha

WHO – World Health Organization

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Chapter 01

Introduction

1.0. Introduction

1.1.1 Definition and Fundamental Concept

Inflammation is a highly complex, coordinated, and adaptive biological response of vascularized tissues to harmful stimuli such as pathogens, damaged cells, toxic compounds, or physical injury. It serves as a protective mechanism designed to eliminate the initial cause of cellular injury, remove necrotic tissue, and initiate the healing process. This intricate response involves the interaction of multiple components including immune cells, blood vessels, extracellular matrix proteins, and a wide range of molecular mediators such as cytokines, chemokines, and growth factors *(*Medzhitov *et al., 2010).* From a physiological perspective, inflammation is a critical component of innate immunity and represents the body's first line of defense against infection and injury. It is a dynamic process that evolves through tightly regulated steps to ensure effective protection while minimizing collateral tissue damage. However, dysregulation of inflammatory responses may lead to excessive or chronic inflammation, contributing significantly to the pathogenesis of numerous diseases including autoimmune disorders, metabolic syndrome, cardiovascular diseases, and cancer (Nathan *et al., 2020).*

1.1.2 Historical Evolution of Inflammation Concept

The concept of inflammation has undergone significant evolution over centuries. In ancient times, the Roman physician Celsus described four cardinal signs of inflammation: redness (rubor), heat (calor), swelling (tumor), and pain (dolor). Later, Rudolf Virchow introduced the fifth sign, loss of function (functio laesa), providing a more comprehensive understanding of inflammatory processes *(Ferrero-Miliani et al., 2007).* With the advancement of modern biomedical sciences, inflammation is now recognized not only as a localized response but also as a systemic process involving complex molecular signaling networks. The integration of immunology, molecular biology, and systems biology has significantly enhanced our understanding of inflammation as a multifactorial and highly regulated biological phenomenon *(Hunter et al., 2012).*

1.1.3 Biological Importance and Dual Nature

Inflammation exhibits a dual role in maintaining physiological homeostasis. On one hand, it acts as a protective mechanism by eliminating invading pathogens, clearing damaged cells, and promoting tissue repair. On the other hand, excessive, prolonged, or uncontrolled inflammation can lead to tissue destruction and contribute to chronic disease development *(Furman et al., 2019).* This dual nature has led to the characterization of inflammation as a "double-edged sword." While acute inflammation is beneficial and essential for survival, chronic inflammation is detrimental and associated with diseases such as rheumatoid arthritis, diabetes mellitus, atherosclerosis, and neurodegenerative disorders *(Nathan et al., 2020).*

1.1.4 Phases of Inflammatory Response

The inflammatory response is a highly orchestrated process consisting of three major phases:

Initiation Phase:

This phase begins with the recognition of harmful stimuli by pattern recognition receptors (PRRs) present on immune cells such as macrophages and dendritic cells. These receptors detect pathogen-associated molecular patterns (PAMPs) and damage-associated molecular patterns (DAMPs), leading to the activation of intracellular signaling pathways and the release of inflammatory mediators *(Chen et al., 2018).*

Amplification Phase:

During this phase, there is a rapid recruitment of immune cells including neutrophils and monocytes to the site of injury. This is accompanied by the release of pro-inflammatory cytokines such as TNF-α, IL-1β, and IL-6, which amplify the inflammatory response and promote further cell recruitment and activation *(Serhan et al., 2014).*

Resolution Phase:

The resolution phase involves the termination of the inflammatory response and restoration of tissue homeostasis. Anti-inflammatory mediators such as lipoxins, resolvins, and cytokines like IL-10 play a crucial role in this phase. Failure of resolution leads to chronic inflammation *(Serhan et al., 2014).*

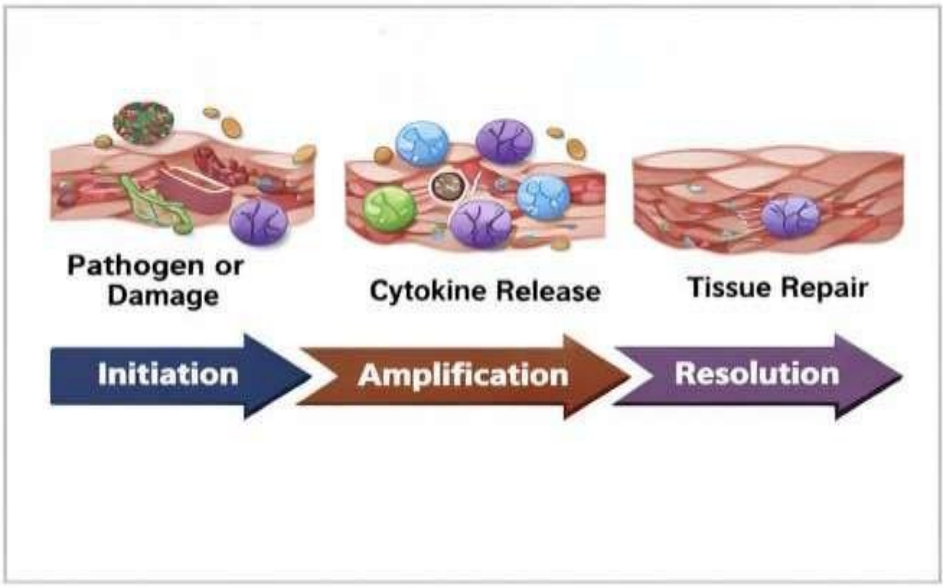


Figure 1: Phases of Inflammation

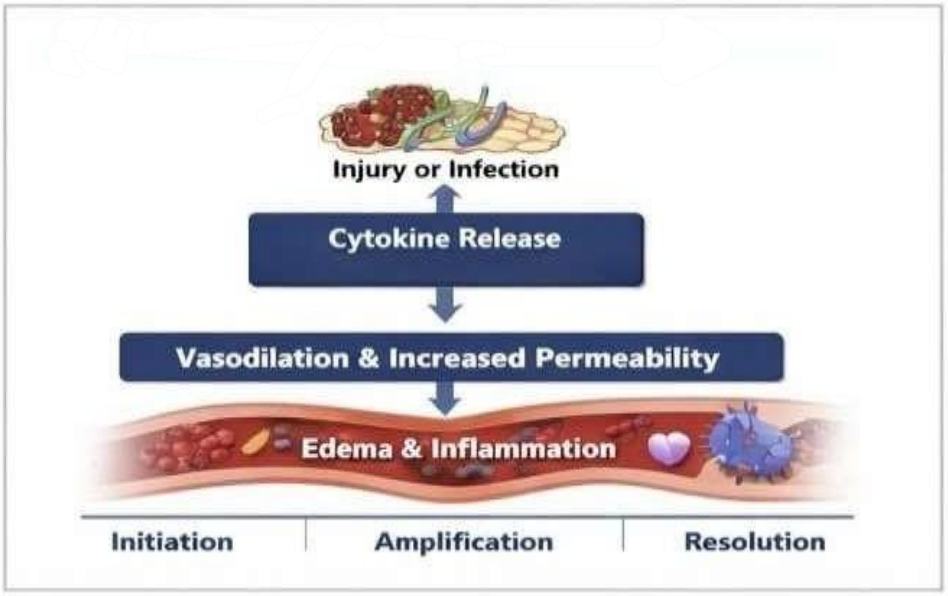


Figure 2: Mechanism of Inflammation

1.1.5 Cellular Components of Inflammation

Inflammation involves a variety of cellular components:

Neutrophils: First responders, responsible for phagocytosis and release of reactive oxygen species

Macrophages: Central regulators of inflammation, producing cytokines and growth

factors

Lymphocytes: Involved in chronic inflammation and adaptive immune responses

Mast Cells: Release histamine and initiate vascular changes

Endothelial Cells: Regulate vascular permeability and leukocyte adhesion

These cells interact dynamically to regulate the inflammatory cascade *(Nathan et al., 2020).*

1.1.6 Molecular Mediators and Their Roles

Inflammation is mediated by a complex network of chemical mediators:

- Cytokines
- TNF-α
- IL-1β
- IL-6

These are key regulators of inflammation and immune responses.

- Eicosanoids
- Prostaglandins
- Leukotrienes

Derived from arachidonic acid, these mediators regulate pain, fever, and vascular changes.

Other Mediators:

- Histamine.
- Nitric oxide.
- Bradykinin.

These molecules contribute to vascular permeability and pain sensation *(Chen et al., 2018).*

1.1.7 Vascular Events in Inflammation

One of the hallmark features of inflammation is vascular response:

- Vasodilation → increased blood flow (redness and heat)
- Increased permeability → plasma leakage (swelling)
- Endothelial activation → leukocyte adhesion

These changes facilitate the migration of immune cells to the site of injury

(Ferrero-Miliani et al., 2007).

1.1.8 Leukocyte Recruitment and Migration

- Leukocyte migration is a multistep process:
- Margination
- Rolling (mediated by selectins)
- Adhesion (mediated by integrins)
- Transmigration

This process ensures targeted delivery of immune cells to inflamed tissues *(Ley et al., 2007).*

1.1.9 Role of Signaling Pathways

Several intracellular signaling pathways regulate inflammation:

- NF-κB pathway
- MAPK pathway
- JAK-STAT pathway
- NLRP3 inflammasome

These pathways control gene expression and cytokine production *(Lawrence et al., 2009).*

1.1.10 Inflammation and Oxidative Stress

Oxidative stress plays a critical role in inflammation. Reactive oxygen species (ROS) generated during inflammation can damage cellular components and amplify inflammatory responses. Antioxidant systems such as SOD, CAT, and GSH help maintain redox balance *(Biswas et al., 2016).*

1.1.11 Systemic Effects of Inflammation

Inflammation can produce systemic effects such as:

- Fever
- Leukocytosis
- Acute phase protein production

These responses are mediated by cytokines and reflect the body's systemic reaction to local injury *(Furman et al., 2019).*

1.1.12 Inflammation in Disease Pathogenesis

Chronic inflammation is implicated in numerous diseases including:

- Rheumatoid arthritis
- Diabetes mellitus
- Cardiovascular diseases
- Alzheimer's disease

Persistent activation of inflammatory pathways contributes to disease progression and complications *(Hunter et al., 2012).*

1.2 Types of Inflammation

1.2.1 Introduction to Types of Inflammation

Inflammation is broadly classified into acute and chronic types based on the duration, nature of response, and underlying mechanisms involved in the inflammatory process. This classification plays a crucial role in understanding disease progression, pathological changes, and therapeutic interventions. Acute inflammation represents the immediate and early response to tissue injury, whereas chronic inflammation is characterized by prolonged and persistent immune activation that may lead to tissue damage and fibrosis *(Nathan et al., 2020).*

The distinction between acute and chronic inflammation is not solely dependent on time duration but also involves differences in cellular composition, mediator profiles, and biological outcomes. Acute inflammation is typically dominated by neutrophils, while chronic inflammation involves macrophages, lymphocytes, and plasma cells. Understanding these variations is essential for designing targeted treatment strategies *(Chen et al., 2018).*

1.2.2 Acute Inflammation

Acute inflammation is a rapid, short-term response that occurs immediately after tissue injury or infection. It usually lasts for a few hours to a few days and is characterized by the exudation of plasma proteins and the migration of leukocytes, predominantly neutrophils, to the site of injury. The primary objective of acute inflammation is to eliminate harmful stimuli and initiate tissue repair *(Ferrero-Miliani et al., 2007).* The hallmark features of acute inflammation include redness, heat, swelling, pain, and loss of function. These clinical manifestations are a result

of vascular changes and mediator release during the inflammatory response *(Nathan et al., 2020).*

1.2.2.1 Vascular Changes in Acute Inflammation

The initial phase of acute inflammation involves significant vascular changes, including vasodilation and increased vascular permeability. Vasodilation results in increased blood flow to the affected area, leading to redness and heat. Increased vascular permeability allows plasma proteins and leukocytes to move from the bloodstream into the surrounding tissues, resulting in swelling *(Ferrero-Miliani et al., 2007).* Endothelial cells play a crucial role in regulating these vascular changes by expressing adhesion molecules that facilitate leukocyte attachment and migration.

1.2.2.2 Cellular Events in Acute Inflammation

Cellular events in acute inflammation involve the recruitment and activation of leukocytes. Neutrophils are the first cells to arrive at the site of injury. The process of leukocyte migration includes margination, rolling, adhesion, and transmigration through the endothelial lining of blood vessels *(Ley et al., 2007).* Once at the site of injury, neutrophils perform phagocytosis and release reactive oxygen species (ROS) and enzymes to eliminate pathogens. This process is essential for host defense but may also contribute to tissue damage if excessive.

1.2.2.3 Chemical Mediators in Acute Inflammation

Chemical mediators play a central role in regulating the inflammatory response. These include histamine, prostaglandins, leukotrienes, and cytokines such as TNF-α, IL-1β, and IL-6. Histamine is responsible for vasodilation and increased vascular permeability, while prostaglandins contribute to pain and fever. Leukotrienes facilitate leukocyte recruitment, and cytokines amplify the inflammatory response *(Chen et al., 2018).*

1.2.2.4 Outcomes of Acute Inflammation

The outcome of acute inflammation depends on the severity and duration of

the response. Possible outcomes include complete resolution, healing by fibrosis, abscess formation, or progression to chronic inflammation. Resolution occurs when the inflammatory response successfully eliminates the harmful stimulus and restores normal tissue structure and function *(Serhan et al., 2014).*

1.2.3 Chronic Inflammation

Chronic inflammation is a prolonged inflammatory response that persists for weeks, months, or even years. It is characterized by the presence of macrophages, lymphocytes, and plasma cells, along with tissue destruction and attempts at repair. Chronic inflammation is often associated with persistent infections, autoimmune diseases, or prolonged exposure to toxic agents *(Hunter et al., 2012).*

Unlike acute inflammation, chronic inflammation involves continuous activation of immune pathways, leading to progressive tissue damage and fibrosis *(Furman et al., 2019).*

1.2.3.1 Causes of Chronic Inflammation

Chronic inflammation may arise due to several factors, including persistent microbial infections, autoimmune reactions, prolonged exposure to irritants, and metabolic disorders. These factors lead to sustained immune activation and continuous release of inflammatory mediators *(Hunter et al., 2012).*

1.2.3.2 Cellular Composition in Chronic Inflammation

The cellular composition of chronic inflammation includes macrophages, lymphocytes, and plasma cells. Macrophages play a central role by producing cytokines and growth factors, while lymphocytes contribute to adaptive immune responses. Plasma cells are responsible for antibody production *(Nathan et al., 2020).*

1.2.3.3 Mechanisms of Tissue Damage

Tissue damage in chronic inflammation occurs due to continuous release of inflammatory mediators, reactive oxygen species, and proteolytic enzymes. These factors lead to cell injury, fibrosis, and organ dysfunction *(Biswas et al., 2016).*

1.2.3.4 Granulomatous Inflammation

Granulomatous inflammation is a specialized form of chronic inflammation characterized by the formation of granulomas, which are organized collections of macrophages. This type of inflammation is commonly observed in diseases such as tuberculosis and sarcoidosis *(Hunter et al., 2012).*

1.2.4 Comparison Between Acute and Chronic Inflammation

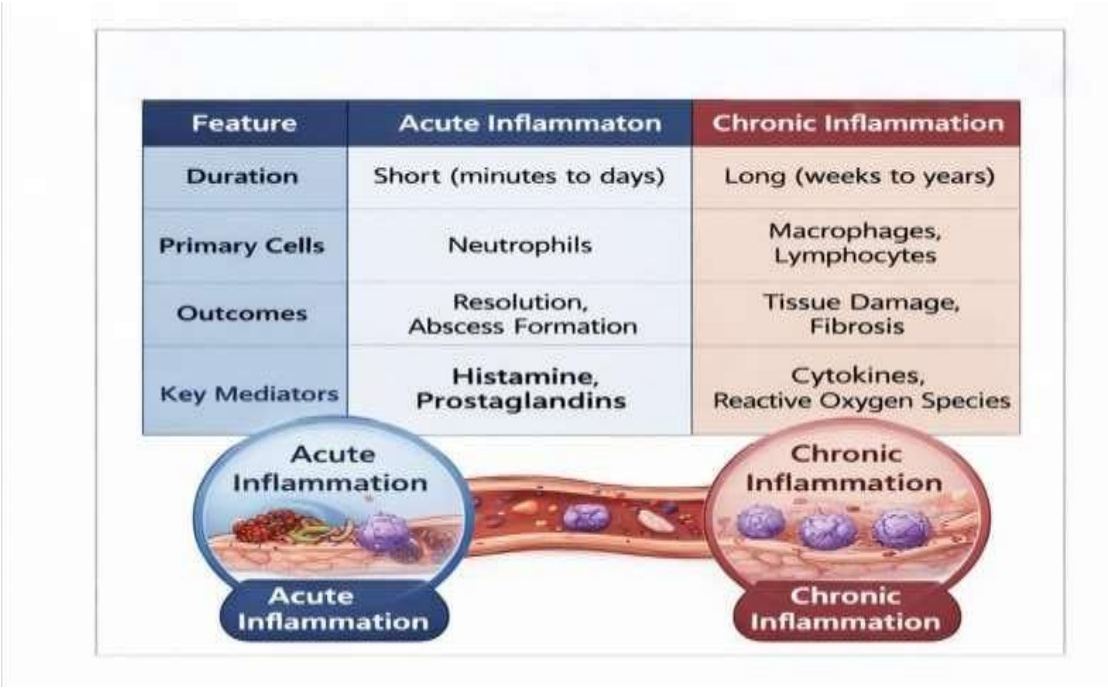
Acute and chronic inflammation differ significantly in terms of duration, cellular components, mediators, and outcomes. Acute inflammation is short-term and primarily involves neutrophils, whereas chronic inflammation is long-lasting and involves macrophages and lymphocytes. Acute inflammation generally results in resolution, while chronic inflammation leads to tissue damage and fibrosis.

1.2.5 Subtypes of Inflammation

Inflammation can also be classified into different subtypes based on the nature of exudate and tissue response. These include serous inflammation, characterized by fluid accumulation; fibrinous inflammation, involving fibrin deposition; and suppurative inflammation, marked by pus formation due to neutrophil accumulation.

1.2.6 Clinical Significance

Understanding the types of inflammation is essential for clinical diagnosis and treatment. Acute inflammation requires immediate intervention to eliminate harmful stimuli, while chronic inflammation requires long-term management strategies to prevent tissue damage and disease progression *(Furman et al., 2019).*



Feature	Acute Inflammaton	Chronic Inflammation
Duration	Short (minutes to days)	Long (weeks to years)
Primary Cells	Neutrophils	Macrophages, Lymphocytes
Outcomes	Resolution, Abscess Formation	Tissue Damage, Fibrosis
Key Mediators	Histamine, Prostaglandins	Cytokines, Reactive Oxygen Species

Figure 3: Comparison of Acute and Chronic Inflammation

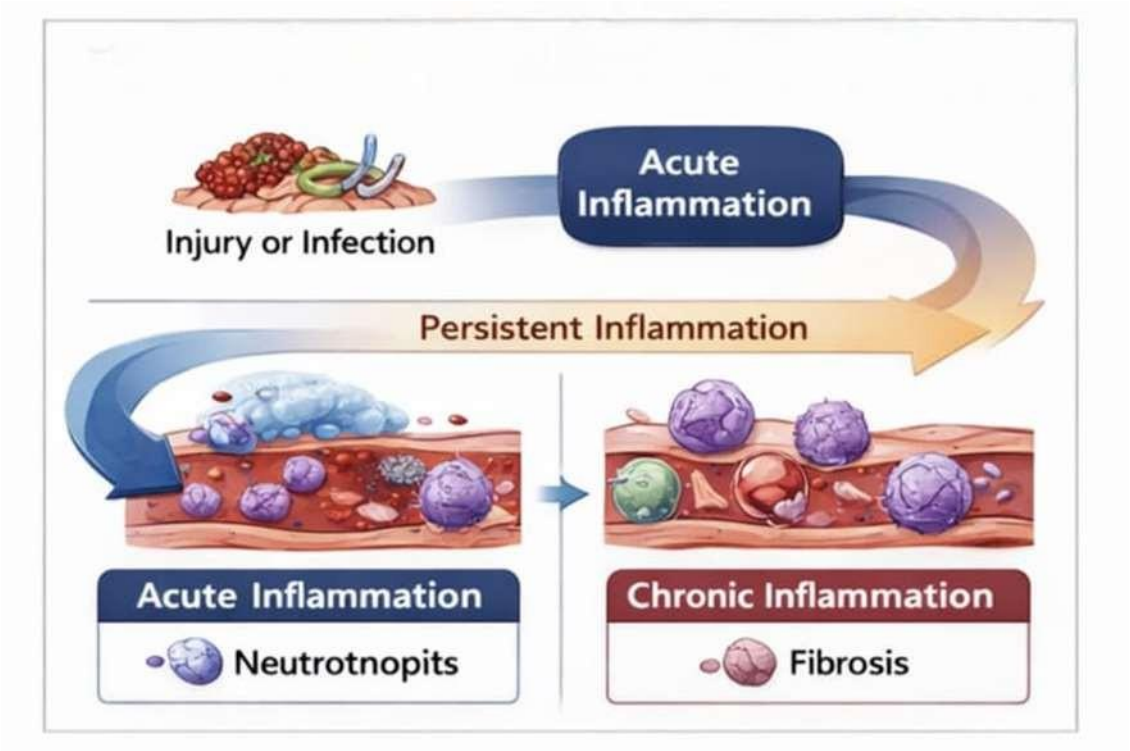


Figure 4: Progression from Acute to Chronic Inflammation

1.3 Pathophysiology of Inflammation

1.3.1 Introduction to Pathophysiology of Inflammation

The pathophysiology of inflammation involves a complex interplay of cellular, molecular, and biochemical events that occur in response to tissue injury or infection. It represents a highly coordinated defense mechanism aimed at eliminating harmful stimuli and initiating tissue repair. However, dysregulation of these processes can lead to chronic inflammation and disease progression *(Medzhitov et al., 2010).* Inflammation is mediated by a network of signaling pathways, immune cells, and chemical mediators that work together to maintain homeostasis. Understanding the pathophysiological mechanisms of inflammation is essential for developing effective therapeutic strategies (*Nathan et al., 2020).*

1.3.2 Recognition of Inflammatory Stimuli

The inflammatory response begins with the recognition of harmful stimuli by pattern recognition receptors (PRRs) such as Toll-like receptors (TLRs) and NOD-like receptors (NLRs). These receptors detect pathogen-associated molecular patterns (PAMPs) and damage-associated molecular patterns (DAMPs), triggering intracellular signaling pathways *(Chen et al., 2018).* Activation of these receptors leads to the production of pro-inflammatory cytokines and chemokines, initiating the inflammatory cascade.

1.3.3 Activation of Signaling Pathways

Several intracellular signaling pathways play a crucial role in inflammation:

- NF-κB pathway → regulates cytokine production
- MAPK pathway → controls cellular responses
- JAK-STAT pathway → mediates immune signaling
- NLRP3 inflammasome → activates IL-1β

These pathways coordinate gene expression and inflammatory mediator release *(Lawrence et al., 2009).*

1.3.4 Role of Cytokines and Chemokines

Cytokines and chemokines are key mediators in inflammation. Pro-

inflammatory cytokines such as TNF-α, IL-1β, and IL-6 promote inflammation, while anti-inflammatory cytokines like IL-10 regulate and suppress the response. Chemokines guide the migration of immune cells to the site of injury, ensuring targeted immune response *(Nathan et al., 2020).*

1.3.5 Cellular Events in Pathophysiology

Inflammation involves various immune cells:

- Neutrophils → early response
- Macrophages → cytokine production
- Lymphocytes → chronic inflammation
- These cells interact to regulate inflammatory processes and tissue repair.

1.3.6 Vascular Events in Inflammation

Vascular changes are central to inflammation and include:

- Vasodilation
- Increased vascular permeability
- Endothelial activation

These changes facilitate immune cell migration and plasma protein leakage *(Ferrero-Miliani et al., 2007).*

1.3.7 Role of Reactive Oxygen Species (ROS)

Reactive oxygen species are generated during inflammation and play a dual role. While they help in pathogen destruction, excessive ROS can cause oxidative damage to cellular components such as lipids, proteins, and DNA *(Biswas et al., 2016*

1.3.8 Enzymatic Mediators and Tissue Damage

Enzymes such as proteases, cyclooxygenase (COX), and lipoxygenase (LOX) contribute to inflammation by producing mediators and degrading extracellular matrix components. These processes can lead to tissue damage if not regulated.

1.3.9 Resolution of Inflammation

Resolution is an active process involving anti-inflammatory mediators such as lipoxins, resolvins, and cytokines. It restores tissue homeostasis and prevents chronic inflammation *(Serhan et al., 2014).*

1.3.10 Failure of Resolution and Chronic Inflammation

Failure of resolution leads to persistent inflammation characterized by continuous immune activation, tissue destruction, and fibrosis. This contributes to the development of chronic diseases.

1.3.11 Systemic Pathophysiological Effects

Inflammation can have systemic effects including fever, leukocytosis, and acute phase protein production. These responses are mediated by cytokines and reflect systemic involvement *(Furman et al., 2019).*

1.3.12 Clinical Implications

Understanding the pathophysiology of inflammation is crucial for the development of anti-inflammatory drugs and therapeutic interventions. Targeting specific pathways can help in managing inflammatory diseases effectively.

1.4 Etiology of Inflammation

1.4.1 Introduction

Etiology of inflammation refers to the underlying causes and contributing factors responsible for initiating and sustaining the inflammatory response. Inflammation is triggered by a wide variety of endogenous and exogenous stimuli, including infections, physical injury, chemical agents, immune reactions, and metabolic disturbances. These causative factors activate complex molecular pathways that lead to the recruitment of immune cells, release of inflammatory mediators, and subsequent tissue responses *(Medzhitov et al., 2010).*

The etiology of inflammation is multifactorial in nature and involves both external and internal factors. Understanding these causative mechanisms is essential for identifying disease progression and designing effective therapeutic interventions.

Based on the nature of stimuli, the causes of inflammation can be broadly classified into infectious and non-infectious categories.

1.4.2 Infectious Causes

Infectious agents are among the most common causes of inflammation and include bacteria, viruses, fungi, and parasites. These pathogens trigger inflammation by interacting with pattern recognition receptors such as Toll-like receptors (TLRs) and NOD-like receptors (NLRs) present on immune cells. This interaction leads to activation of intracellular signaling pathways and production of pro-inflammatory cytokines *(Chen et al., 2018).* Bacterial infections typically induce acute inflammation characterized by neutrophil infiltration, whereas viral infections often lead to chronic inflammatory responses involving lymphocytes. Fungal and parasitic infections can also cause persistent inflammatory responses depending on the nature of the pathogen.

1.4.3 Physical Causes

Physical factors such as mechanical trauma, burns, radiation, and extreme temperatures can cause direct tissue injury and initiate inflammation. These injuries lead to the release of damage-associated molecular patterns (DAMPs), which activate immune cells and trigger the inflammatory cascade.

Radiation exposure, particularly ionizing radiation, can cause cellular damage by inducing DNA breaks and oxidative stress, leading to inflammation and tissue injury.

1.4.4 Chemical Causes

Chemical agents including environmental pollutants, toxins, drugs, and irritants can initiate inflammation by causing cellular damage and oxidative stress. These substances may disrupt cellular membranes, denature proteins, and activate inflammatory signaling pathways such as NF-κB and MAPK.

Chronic exposure to harmful chemicals can result in prolonged inflammation and contribute to the development of diseases such as cancer and respiratory disorders.

1.4.5 Immunological Causes

Inflammation can arise due to dysregulation of the immune system. Autoimmune diseases occur when the immune system mistakenly attacks self-tissues, leading to chronic inflammation. Hypersensitivity reactions and allergic responses are also important immunological causes of inflammation *(Nathan et al., 2020).* These conditions involve excessive activation of immune cells and overproduction of cytokines, resulting in tissue damage and disease progression.

1.4.6 Metabolic and Lifestyle Factors

Metabolic disorders such as obesity, diabetes mellitus, and hyperlipidemia are strongly associated with chronic low-grade inflammation. Adipose tissue produces inflammatory cytokines that contribute to systemic inflammation.

Lifestyle factors including smoking, alcohol consumption, poor diet, and psychological stress further exacerbate inflammatory responses and increase the risk of chronic diseases *(Furman et al., 2019).*

1.4.7 Genetic Factors

Genetic predisposition plays a significant role in determining an individual's susceptibility to inflammatory diseases. Mutations or polymorphisms in genes regulating immune responses, cytokine production, and signaling pathways can alter inflammatory processes.

Genetic factors may influence both the intensity and duration of inflammation, thereby contributing to disease susceptibility and progression.

1.4.8 Environmental Factors

Environmental exposure to pollutants, allergens, and toxic substances can trigger inflammation. Air pollution, occupational hazards, and exposure to allergens are major contributors to inflammatory diseases such as asthma and cardiovascular disorders. Long-term exposure to environmental risk factors can lead to chronic inflammation and associated complications.

Parameter	Description	Key Mediators	Cells| Outcome
Definition|	Protective biological response to harmful stimuli such as pathogens, injury, or toxins	Immune cells (macrophages, neutrophils)	Initiation of defense mechanism
Types of Inflammation	Acute (short-term) and Chronic (long-term) inflammation	Acute: neutrophils; Chronic: macrophages, lymphocytes	Tissue repair or chronic damage
Causes	Infection, tissue injury, oxidative stress, autoimmune disorders	Pathogens, ROS, cytokines	Activation of inflammatory response
Key Mediators	Chemical mediators involved in inflammation	TNF-α, IL-1β, IL-6, prostaglandins, leukotrienes	Amplification of inflammation
Pathophysiology	Activation of signaling pathways and release of mediators	NF-κB, MAPK, COX-2, iNOS pathways	Edema, pain, redness, tissue damage

Table 1: Overview of Inflammation

Flowchart: Etiology of Inflammation

Etiology of Inflammation

↓

Infectious Causes (Bacteria, Viruses, Fungi, Parasites)

↓

Physical Causes (Trauma, Burns, Radiation, Temperature)

↓

Chemical Causes (Toxins, Drugs, Pollutants)

↓

Immunological Causes (Autoimmune Diseases, Allergy, Hypersensitivity)

↓

Metabolic Factors (Obesity, Diabetes, Lifestyle)

↓

Genetic Factors (Gene Mutations, Susceptibility)

↓

Environmental Factors (Pollution, Allergens, Toxins)

Chapter 02

Literature Review

2.0 Literature Review

The literature review serves as a fundamental component of any scientific research, providing a comprehensive and critical analysis of previously published studies related to the research topic. In the context of the present study, which focuses on the evaluation of anti-inflammatory activity of curcumin and nimbidin in combination, the literature review aims to consolidate existing knowledge on inflammation, its underlying molecular mechanisms, and the therapeutic potential of natural bioactive compounds. It establishes a scientific foundation for the study by identifying gaps in current research and justifying the rationale for investigating combination therapy using phytoconstituents *(Gupta et al., 2021).*

Inflammation is a highly complex biological response involving multiple cellular and molecular processes, including the activation of immune cells, release of inflammatory mediators, and regulation of signaling pathways. Over the past decade, extensive research has been conducted to understand the intricate mechanisms governing inflammation and its role in the progression of various chronic diseases such as arthritis, cardiovascular disorders, neurodegenerative diseases, and cancer. These studies have highlighted the importance of targeting key inflammatory pathways such as nuclear factor-kappa B (NF-κB), mitogen-activated protein kinase (MAPK), and NLRP3 inflammasome for effective therapeutic intervention *(Medzhitov, 2021).*

Recent advancements in pharmacological research (2020–2025) have emphasized the limitations of conventional anti-inflammatory drugs, including non-steroidal anti-inflammatory drugs (NSAIDs) and corticosteroids. Although these agents are effective in reducing inflammation, their long-term use is associated with significant adverse effects such as gastrointestinal toxicity, cardiovascular risks, and immunosuppression. These limitations have driven the scientific community to explore safer and more effective alternatives, particularly those derived from natural sources with multitargeted mechanisms of action *(Vane and Botting, 2021).*

Among natural compounds, curcumin, a polyphenolic compound derived from Curcuma longa, has gained significant attention due to its potent anti-inflammatory, antioxidant, and immunomodulatory properties. Curcumin exerts its effects by modulating various signaling pathways, including inhibition of NF-κB activation, suppression of pro-inflammatory cytokines such as tumor necrosis factor-alpha (TNF-

α) and interleukin-6 (IL-6), and downregulation of cyclooxygenase-2 (COX-2) expression. Numerous in vitro and in vivo studies have demonstrated the therapeutic potential of curcumin in managing inflammatory conditions *(Hewlings and Kalman, 2017; Zhang et al., 2022).*

Similarly, nimbidin, a bioactive compound isolated from Azadirachta indica (Neem), has been reported to possess significant anti-inflammatory activity. Nimbidin acts by inhibiting the release of histamine and prostaglandins, reducing leukocyte migration, and suppressing the production of inflammatory mediators. It also exhibits antioxidant properties, thereby reducing oxidative stress associated with inflammation. Previous studies have demonstrated its efficacy in various experimental models of inflammation, suggesting its potential as a therapeutic agent *(Biswas et al., 2002; Sharma et al., 2021).*

Despite the individual efficacy of curcumin and nimbidin, recent research trends have focused on the concept of combination therapy, which involves the use of two or more agents to achieve enhanced therapeutic effects. Combination therapy offers several advantages, including synergistic action, reduced dosage requirements, and minimized side effects. In the case of curcumin and nimbidin, their complementary mechanisms of action provide a strong rationale for their combined use in the management of inflammation. While curcumin primarily targets intracellular signaling pathways such as NF-κB and MAPK, nimbidin acts on extracellular mediators such as histamine and prostaglandins, thereby providing a broader spectrum of anti-inflammatory activity (*Patel et al., 2023).*

Furthermore, emerging approaches such as network pharmacology have provided deeper insights into the multi-target mechanisms of phytoconstituents. These approaches highlight the ability of natural compounds to modulate multiple signaling pathways simultaneously, making them highly effective in treating complex diseases like inflammation. The integration of curcumin and nimbidin into a combination therapy aligns with this paradigm, as it enables simultaneous targeting of multiple inflammatory mediators and pathways, thereby enhancing therapeutic efficacy (*Li et al., 2022).*

Another important aspect highlighted in recent literature is the role of oxidative stress in inflammation. Reactive oxygen species (ROS) generated during inflammatory

responses contribute to cellular damage and activation of pro-inflammatory signaling pathways. Both curcumin and nimbidin have been reported to exhibit strong antioxidant properties, which further support their role in reducing inflammation. By scavenging free radicals and enhancing the activity of antioxidant enzymes, these compounds help in maintaining cellular homeostasis and preventing tissue damage *(Reuter et al., 2010; Kumar et al., 2023).*

Despite significant progress in understanding the pharmacological properties of curcumin and nimbidin, there is limited research focusing on their combined anti-inflammatory effects. This highlights a critical research gap and underscores the need for systematic investigation of their synergistic potential. The present study is therefore designed to evaluate the anti-inflammatory activity of curcumin and nimbidin in combination using suitable experimental models, thereby contributing to the development of safer and more effective therapeutic strategies.

In conclusion, the literature review provides a detailed understanding of inflammation and the therapeutic potential of curcumin and nimbidin. It establishes a strong scientific basis for the present study and highlights the importance of exploring combination therapy as a promising approach in the management of inflammatory disorders.

2.1 Inflammation

Inflammation is a complex biological response of vascular tissues to harmful stimuli such as pathogens, damaged cells, or irritants. It serves as a protective mechanism intended to eliminate the initial cause of cell injury, remove necrotic cells and tissues, and initiate tissue repair. The inflammatory response involves a coordinated interaction between immune cells, blood vessels, and molecular mediators. This process is tightly regulated and essential for maintaining homeostasis *(Medzhitov et al., 2021).* Inflammation can be beneficial or harmful depending on its duration and intensity. Acute inflammation is a short-term response that facilitates healing, whereas chronic inflammation is prolonged and contributes to the development of various pathological conditions such as arthritis, cardiovascular diseases, diabetes, and cancer. Recent research has highlighted the role of signaling pathways such as NF-κB, MAPK, and NLRP3 inflammasome in regulating inflammatory responses *(Nathan et al., 2020).*

2.1.1 Types of Inflammation

Inflammation is broadly classified into acute and chronic inflammation based on duration, cellular involvement, and outcome.

2.1.1.1 Acute Inflammation

Acute inflammation is the immediate and early response to tissue injury or infection. It is characterized by rapid onset and short duration, typically lasting from a few hours to a few days. The primary objective is to eliminate the injurious agent and initiate the healing process. The hallmark features of acute inflammation include vasodilation, increased vascular permeability, leukocyte migration, and release of inflammatory mediators. Neutrophils are the predominant cells involved in this phase

Table 2: Characteristics of Acute Inflammation

Features	Description
Onset	Rapid
Duration	Short (hours to days)
Primary cells	Neutrophils
Outcomes	Resolution or Healing

2.1.1.2 Chronic Inflammation

Chronic inflammation is a prolonged inflammatory response that may persist for weeks, months, or years. It results from the failure to eliminate the causative agent, persistent immune activation, or autoimmune reactions. Chronic inflammation is characterized by infiltration of macrophages, lymphocytes, and plasma cells. It is associated with tissue destruction and attempts at repair, including fibrosis and angiogenesis. Chronic inflammatory conditions are implicated in diseases such as rheumatoid arthritis, cancer, and neurodegenerative disorders.

Table 3: Difference Between Acute and Chronic Inflammation

Features	Acute	Chronic
Duration	Short	Long
Cells involved	Neutrophils	Macrophages Lymphocytes
Outcomes	Healing	Tissue damage

2.1.2 Cellular Events in Inflammation

The cellular phase of inflammation involves the migration of leukocytes from the bloodstream to the site of injury. This process is essential for the elimination of pathogens and damaged tissues.

The steps involved in leukocyte migration include margination, rolling, adhesion, diapedesis, and chemotaxis. These processes are mediated by adhesion molecules such as selectins and integrins. Chemotactic factors guide leukocytes to the site of inflammation where they perform phagocytosis and release inflammatory mediators.

2.1.3 Chemical Mediators of Inflammation

Inflammation is regulated by various chemical mediators derived from plasma proteins and cells. These mediators are responsible for vascular changes, leukocyte recruitment, and amplification of the inflammatory response.

2.1.3.1 Cytokines

Cytokines are small signaling proteins that regulate immune and inflammatory responses. Pro-inflammatory cytokines such as TNF-α, IL-1β, and IL-6 play a key role in initiating and sustaining inflammation.

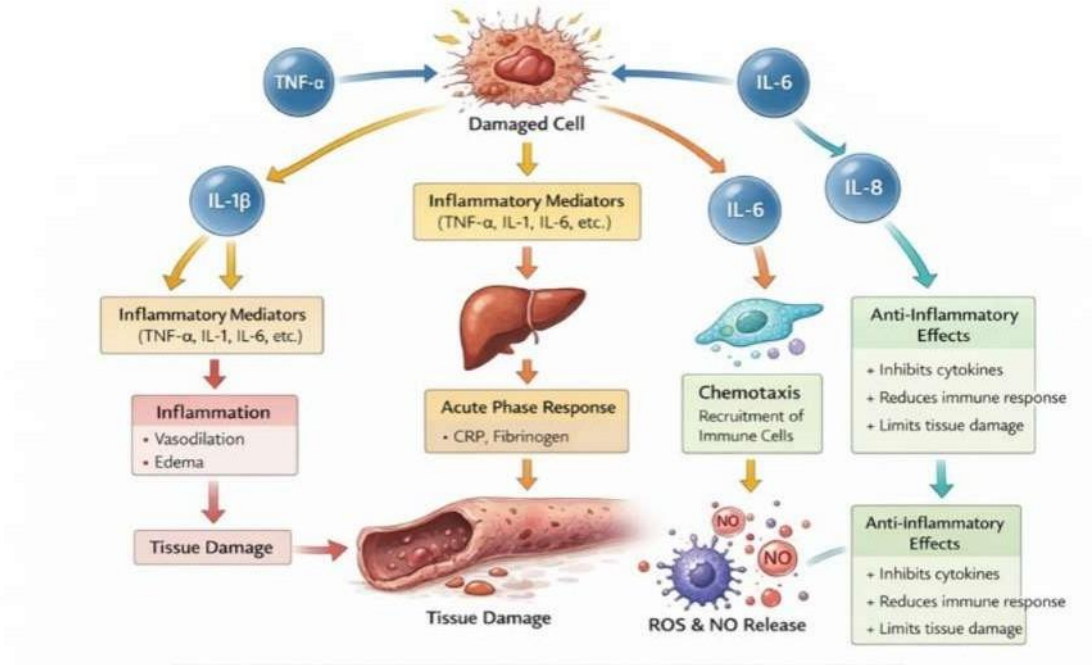


Figure 5: Inflammatory Response Mediated by Cytokines

2.1.3.2 Prostaglandins

Prostaglandins are lipid mediators produced from arachidonic acid via the cyclooxygenase (COX) pathway. They are responsible for pain, fever, and vasodilation.

2.1.3.3 Leukotrienes

Leukotrienes are produced via the lipoxygenase pathway and are involved in bronchoconstriction, increased vascular permeability, and leukocyte recruitment.

2.1.3.4 Histamine and Serotonin

Histamine is released from mast cells and causes vasodilation and increased vascular permeability. Serotonin also contributes to vascular changes during inflammation.

Table 4: Major Inflammatory Mediators

Mediator	Source	Function
TNF-α	Macrophages	Cytokine activation
IL-6	Immune cells	Inflammation regulation
Prostaglandins	COX pathway	Pain and fever
Leukotrienes	LOX pathway	Bronchoconstriction

2.1.4 Molecular Mechanisms of Inflammation

Inflammation is regulated by several intracellular signaling pathways that control gene expression and mediator production.

2.1.4.1 NF-κB Signaling Pathway

NF-κB is a key transcription factor that regulates the expression of inflammatory genes. Activation of NF-κB leads to increased production of cytokines, chemokines, and adhesion molecules.

2.1.4.2 MAPK Pathway

MAPK signaling pathways, including ERK, JNK, and p38, play a crucial role in transmitting inflammatory signals and regulating cellular responses.

2.1.4.3 NLRP3 Inflammasome

The NLRP3 inflammasome is involved in the activation of caspase-1 and production of pro-inflammatory cytokines such as IL-1β and IL-18.

2.1.5 Role of Oxidative Stress in Inflammation

Oxidative stress plays a significant role in inflammation through the generation of reactive oxygen species (ROS). These molecules cause cellular damage and activate inflammatory pathways.

Table 5: Reactive Oxygen Species and Their Effects

ROS	Effect
Superoxide	Cellular damage
Hydrogen peroxide	Signaling molecule
Hydroxyl radicals	DNA damage

2.1.6 Resolution of Inflammation

Resolution of inflammation is an active and regulated process that leads to the restoration of tissue homeostasis. It involves the clearance of inflammatory cells and mediators, followed by tissue repair and regeneration

2.2 Arachidonic Acid Pathway in Inflammation

The arachidonic acid pathway is one of the most critical biochemical pathways involved in the regulation of inflammation. It plays a central role in the synthesis of various inflammatory mediators such as prostaglandins, thromboxanes, and leukotrienes, which are responsible for the initiation and progression of inflammatory responses. This pathway serves as a key target for anti-inflammatory drugs, including both synthetic agents and natural compounds such as curcumin and nimbidin *(Ricciotti and FitzGerald, 2011; Calder, 2020).*

Arachidonic acid is a polyunsaturated fatty acid present in the phospholipids of cell membranes. Upon cellular injury or stimulation by inflammatory signals, arachidonic acid is released from membrane phospholipids by the action of phospholipase A2 (PLA2). Once released, it is metabolized through two major enzymatic pathways: the cyclooxygenase (COX) pathway and the lipoxygenase (LOX)

pathway, leading to the production of various pro-inflammatory mediators *(Dennis and Norris, 2015*).

2.2.1 Cyclooxygenase (COX) Pathway

The cyclooxygenase pathway is responsible for the conversion of arachidonic acid into prostaglandins and thromboxanes. This pathway is mediated by cyclooxygenase enzymes, primarily COX-1 and COX-2. COX-1 is constitutively expressed and plays a role in maintaining physiological functions such as gastric mucosal protection and platelet aggregation. In contrast, COX-2 is an inducible enzyme that is upregulated during inflammation and is primarily responsible for the production of pro-inflammatory prostaglandins. Prostaglandins such as PGE2 contribute to vasodilation, increased vascular permeability, pain, and fever, which are hallmark features of inflammation. Thromboxanes are involved in platelet aggregation and vasoconstriction *(Vane and Botting, 2021).* Curcumin has been reported to inhibit COX-2 expression and reduce prostaglandin synthesis, thereby exerting anti-inflammatory effects. Similarly, nimbidin has shown the ability to suppress prostaglandin production, further supporting its role in modulating this pathway *(Hewlings and Kalman, 2017; Sharma et al., 2021).*

Table 6: Difference Between COX-1 and COX-2

Feature	COX-1	COX-2
Expression	Constitutive	Inducible
Function	Physiological	Inflammatory
Role	Gastric protection	Inflammation
Inhibition	Causes side effects	Therapeutic target

2.2.2 Lipoxygenase (LOX) Pathway

The lipoxygenase pathway converts arachidonic acid into leukotrienes and hydroxy eicosatetraenoic acids (HETEs). The enzyme 5-lipoxygenase (5-LOX) plays a major role in this process.

Leukotrienes are potent inflammatory mediators that contribute to bronchoconstriction, increased vascular permeability, and recruitment of leukocytes to the site of

inflammation. They are particularly important in conditions such as asthma and allergic inflammation (*Peters-Golden and Henderson, 2007*).

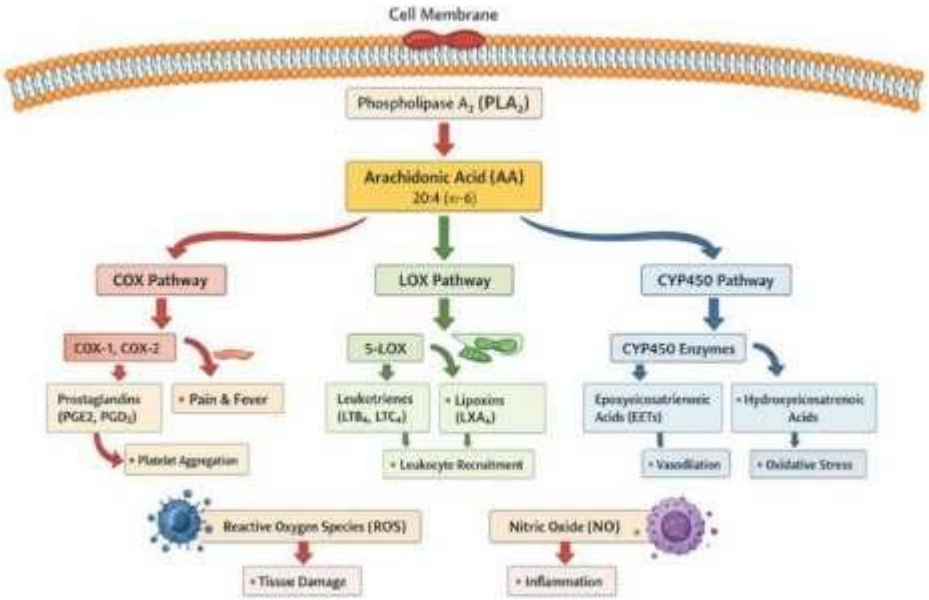


Figure 6: Inflammatory pathways involved in Arachidonic acid metabolism

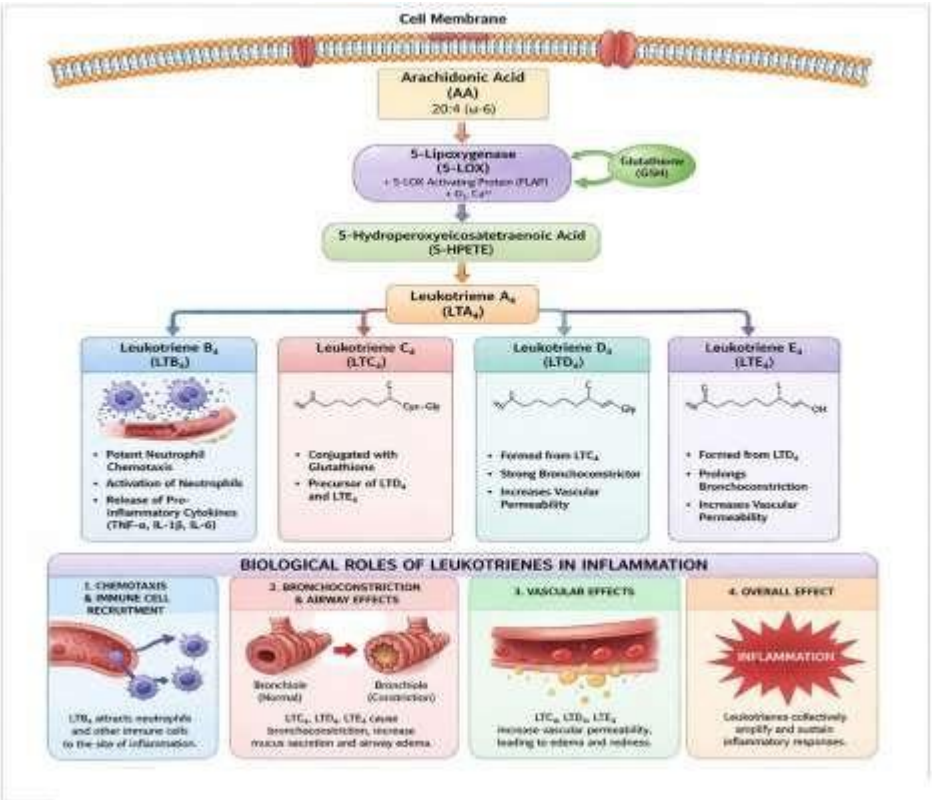


Figure 7: Lipoxygenase pathway showing Leukotrienes synthesis & its role

2.2.3 Role of Arachidonic Acid Metabolites in Inflammation

The metabolites produced through COX and LOX pathways play a significant role in the inflammatory process.

- Prostaglandins → pain, fever, vasodilation
- Thromboxanes → platelet aggregation
- Leukotrienes → bronchoconstriction and chemotaxis

These mediators amplify the inflammatory response and contribute to tissue damage if not regulated properly (*Calder, 2020).*

2.2.4 Therapeutic Targeting of Arachidonic Acid Pathway

The arachidonic acid pathway is a major target for anti-inflammatory therapy. Conventional drugs such as NSAIDs inhibit COX enzymes, while leukotriene inhibitors target the LOX pathway. However, these drugs are associated with adverse effects such as gastrointestinal toxicity and cardiovascular risks. Therefore, natural compounds like curcumin and nimbidin are being explored as safer alternatives due to their ability to modulate multiple targets within the arachidonic acid pathway. Curcumin inhibits both COX-2 and LOX pathways, thereby reducing the production of prostaglandins and leukotrienes. Nimbidin complements this action by inhibiting histamine release and reducing mediator synthesis. The combination of these two compounds provides a synergistic effect by targeting multiple steps in the inflammatory cascade *(Patel et al., 2023).*

2.2.5 Relevance to Present Study

The arachidonic acid pathway is highly relevant to the present study as it represents a central mechanism in inflammation. The ability of curcumin and nimbidin to inhibit key enzymes such as COX and LOX highlights their potential as effective anti-inflammatory agents. Their combined action is expected to provide enhanced therapeutic efficacy by simultaneously reducing multiple inflammatory mediators. This multi-target approach aligns with recent research trends emphasizing the importance of combination therapy in managing complex inflammatory conditions *(Li et al., 2022*).

2.3 Role of Cytokines in Inflammation

Cytokines are small, soluble signaling proteins that play a crucial role in regulating immune and inflammatory responses. They are secreted by various immune cells such as macrophages, lymphocytes, and dendritic cells in response to infection, injury, or immunological stimuli. Cytokines act as key mediators of inflammation by modulating cell communication, immune cell activation, and the release of other inflammatory mediators *(Dinarello, 2007; Tanaka et al., 2014).*

In the inflammatory process, cytokines function in a highly coordinated network to initiate, amplify, and regulate the immune response. Based on their function, cytokines are broadly classified into pro-inflammatory and anti-inflammatory cytokines. An imbalance between these cytokines leads to pathological conditions associated with chronic inflammation *(Zhang and An, 2007).*

2.3.1 Pro-inflammatory Cytokines

Pro-inflammatory cytokines are responsible for initiating and sustaining the inflammatory response. They promote leukocyte recruitment, increase vascular permeability, and stimulate the production of other inflammatory mediators.

2.3.1.1 Tumor Necrosis Factor-alpha (TNF-α)

TNF-α is one of the most important pro-inflammatory cytokines produced primarily by activated macrophages. It plays a central role in inflammation by inducing the expression of adhesion molecules, promoting leukocyte migration, and stimulating the production of other cytokines such as IL-1 and IL-6. TNF-α activates intracellular signaling pathways such as NF-κB and MAPK, leading to the transcription of inflammatory genes. Elevated levels of TNF-α are associated with various inflammatory diseases including rheumatoid arthritis and inflammatory bowel disease (*Bradley, 2008).*

Curcumin has been reported to significantly inhibit TNF-α production by suppressing NF-κB activation. Nimbidin also reduces TNF-α levels by inhibiting mediator release, thereby contributing to its anti-inflammatory effects *(Hewlings and Kalman, 2017; Sharma et al., 2021).*

2.3.1.2 Interleukin-1 beta (IL-1β)

IL-1β is a potent pro-inflammatory cytokine produced by activated macrophages and monocytes. It is involved in fever induction, leukocyte recruitment, and activation of endothelial cells. IL-1β is produced through the activation of the NLRP3 inflammasome, which plays a crucial role in innate immunity. Dysregulation of IL-1β is associated with chronic inflammatory diseases (*Dinarello, 2011).*

2.3.1.3 Interleukin-6 (IL-6)

IL-6 is a multifunctional cytokine that plays a role in both acute and chronic inflammation. It is produced by macrophages, fibroblasts, and endothelial cells.
IL-6 stimulates the production of acute-phase proteins and contributes to the systemic effects of inflammation such as fever. Elevated IL-6 levels are associated with chronic inflammatory conditions. Curcumin reduces IL-6 levels by inhibiting transcription factors, whereas nimbidin modulates IL-6 production through suppression of inflammatory signaling pathways *(Tanaka et al., 2014).*

2.3.2 Anti-inflammatory Cytokines

Anti-inflammatory cytokines play a crucial role in limiting and resolving inflammation. They counteract the effects of pro-inflammatory cytokines and promote tissue repair.

2.3.2.1 Interleukin-10 (IL-10)

IL-10 is a key anti-inflammatory cytokine that suppresses the production of pro-inflammatory cytokines such as TNF-α, IL-1β, and IL-6. It inhibits macrophage activation and promotes immune regulation. Curcumin has been shown to enhance IL-10 production, thereby contributing to its anti-inflammatory effects. Nimbidin also supports anti-inflammatory responses by modulating cytokine balance *(Moore et al., 2001*).

2.3.2.2 Transforming Growth Factor-beta (TGF-β)

TGF-β is involved in immune regulation and tissue repair. It suppresses immune responses and promotes healing processes.

2.3.3 Cytokine Network and Crosstalk

Cytokines interact in a complex network known as the cytokine cascade. This network regulates the balance between pro-inflammatory and anti-inflammatory responses. Activation of pro-inflammatory cytokines leads to amplification of inflammation, whereas anti-inflammatory cytokines act as negative regulators to prevent excessive tissue damage. Dysregulation of this balance results in chronic inflammatory diseases *(Zhang et al., 2022).*

2.3.4 Therapeutic Targeting of Cytokines

Cytokines are important therapeutic targets for anti-inflammatory treatment. Several drugs and natural compounds act by modulating cytokine production. Curcumin inhibits multiple cytokines such as TNF-α, IL-1β, and IL-6, while enhancing anti-inflammatory cytokines like IL-10. Nimbidin complements this effect by reducing mediator release and suppressing immune cell activation. The combination of curcumin and nimbidin provides a synergistic effect by targeting multiple cytokines and restoring immune balance. This multi-target approach is highly beneficial in managing complex inflammatory conditions *(Patel et al., 2023).*

2.3.5 Relevance to Present Study

Cytokines play a central role in the pathogenesis of inflammation and represent key targets for therapeutic intervention. The ability of curcumin and nimbidin to modulate cytokine production highlights their potential as effective anti-inflammatory agents.
The present study focuses on evaluating the combined effect of these compounds in regulating cytokine-mediated inflammation, which may lead to improved therapeutic outcomes and reduced side effects *(Li et al., 2022).*

2.4 Oxidative Stress and Free Radicals in Inflammation

Oxidative stress is a critical factor involved in the initiation and progression of inflammation. It is defined as an imbalance between the production of reactive oxygen species (ROS) and the antioxidant defence mechanisms of the body. Excessive generation of ROS leads to cellular damage, activation of inflammatory pathways, and progression of various chronic diseases *(Reuter et al., 2010; Sies et al., 2020).*

Reactive oxygen species are highly reactive molecules derived from oxygen, including superoxide anion (O_2^-), hydrogen peroxide (H_2O_2), and hydroxyl radicals (•OH). These molecules are generated during normal cellular metabolism, particularly in mitochondria, and are also produced by activated immune cells such as neutrophils and macrophages during inflammation *(Pham-Huy et al., 2008).*

2.4.1 Sources of Reactive Oxygen Species

ROS are generated from both endogenous and exogenous sources.

Endogenous sources:

- Mitochondrial respiration
- NADPH oxidase activity
- Peroxisomal metabolism

Exogenous sources:

- Pollution
- Radiation
- Drugs and toxins

During inflammation, activated neutrophils produce large amounts of ROS through the respiratory burst, which plays a role in pathogen destruction but also contributes to tissue damage (*Mittal et al., 2014).*

Table 7: Sources of ROS

Source Type	Examples
Endogenous	Mitochondria, NADPH oxidase
Exogenous	Pollution, radiation

2.4.2 Role of ROS in Inflammation

ROS play a dual role in biological systems. At low levels, they act as signaling molecules, while at high levels, they cause oxidative damage to lipids, proteins, and DNA.

ROS activate key inflammatory pathways such as:

- NF-κB pathway
- MAPK pathway
- NLRP3 inflammasome

Activation of these pathways leads to increased production of pro-inflammatory cytokines such as TNF-α, IL-1β, and IL-6, thereby amplifying the inflammatory response *(Li et al., 2018).*

2.4.3 Antioxidant defence System

The body possesses a well-developed antioxidant defense system to neutralize ROS and maintain redox balance.

Enzymatic antioxidants:

- Superoxide dismutase (SOD)
- Catalase
- Glutathione peroxidase

Non-enzymatic antioxidants:

- Vitamin C
- Vitamin E
- Glutathione

These antioxidants scavenge free radicals and prevent oxidative damage to cellular components *(Sies et al., 2020*).

Table 8: Antioxidant Défense System

Type	Examples	Function
Enzymatic	SOD, Catalase	ROS neutralization
Non-enzymatic	Vitamin C, E	Free radical scavenging

2.4.4 Oxidative Stress and Chronic Inflammation

Persistent oxidative stress contributes to chronic inflammation and is associated with diseases such as diabetes, cardiovascular disorders, cancer, and neurodegenerative diseases.

ROS-induced damage leads to activation of transcription factors and inflammatory mediators, creating a vicious cycle of oxidative stress and inflammation *(Reuter et al., 2010).*

2.4.5 Role of Curcumin in Oxidative Stress

Curcumin exhibits strong antioxidant properties by scavenging free radicals and enhancing antioxidant enzyme activity. It inhibits ROS generation and suppresses oxidative stress-mediated activation of NF-κB and MAPK pathways. Curcumin also increases levels of endogenous antioxidants such as glutathione and superoxide dismutase, thereby protecting cells from oxidative damage *(Hewlings and Kalman, 2017; Kunnumakkara et al., 2019).*

2.4.6 Role of Nimbidin in Oxidative Stress

Nimbidin also possesses significant antioxidant activity. It reduces lipid peroxidation, enhances antioxidant enzyme activity, and inhibits oxidative stress-induced inflammation. It complements the action of curcumin by targeting different pathways involved in oxidative stress, thereby contributing to a synergistic anti-inflammatory effect *(Sharma et al., 2021).*

2.4.7 Synergistic Role of Curcumin and Nimbidin

The combination of curcumin and nimbidin provides enhanced protection against oxidative stress by targeting multiple pathways simultaneously.

- Curcumin → Direct ROS scavenging + NF-κB inhibition
- Nimbidin → Lipid peroxidation inhibition + enzyme modulation

This synergistic action leads to reduced oxidative damage and suppression of inflammatory responses, making the combination highly effective in the treatment of inflammatory conditions *(Patel et al., 2023).*

2.4.8 Relevance to Present Study

Oxidative stress plays a pivotal role in inflammation and serves as an important therapeutic target. The ability of curcumin and nimbidin to modulate oxidative stress highlights their potential as effective anti-inflammatory agents. The present study focuses on evaluating their combined effect in reducing oxidative stress and inflammation, thereby contributing to the development of safer and more effective therapeutic strategies.

2.5 Curcumin

Curcumin is a natural polyphenolic compound derived from the rhizome of Curcuma longa (turmeric), a widely used medicinal plant belonging to the Zingiberaceae family. It has been extensively studied for its broad spectrum of pharmacological activities, including anti-inflammatory, antioxidant, antimicrobial, anticancer, and neuroprotective effects. Among these, its anti-inflammatory activity has gained significant attention due to its ability to modulate multiple molecular targets involved in inflammation *(Hewlings and Kalman, 2017; Kunnumakkara et al., 2019).* Curcumin is considered a pleiotropic molecule as it interacts with various signaling pathways and transcription factors, making it highly effective in controlling complex inflammatory conditions. Recent studies *(2020–2025)* have demonstrated that curcumin can regulate cytokine production, inhibit inflammatory enzymes, and suppress oxidative stress, thereby providing a multi-targeted therapeutic approach *(Aggarwal et al., 2021; Gupta et al., 2022).*

Figure 8: Chemical structure of Curcumin

2.5.1 Source and Chemical Composition

Curcumin is the principal curcuminoid present in turmeric (Curcuma longa). It constitutes approximately 2–5% of turmeric and is responsible for its characteristic yellow color. Along with curcumin, turmeric contains other curcuminoids such as demethoxycurcumin and bisdemethoxycurcumin. Chemically, curcumin is a diferuloylmethane compound with the molecular formula $C_{21}H_{20}O_6$. It consists of two aromatic ring systems connected by a seven-carbon linker containing α, β-unsaturated carbonyl groups. This structure is responsible for its antioxidant and anti-inflammatory properties *(Nelson et al., 2017).*

Table 9: Chemical Properties of Curcumin

Property	Description
Chemical name	Diferuloylmethane
Molecular formula	$C_{21}H_{20}O_6$
Source	Curcuma longa
Nature	Polyphenolic compound

2.5.2 Pharmacological Activities of Curcumin

Curcumin exhibits a wide range of pharmacological properties due to its ability to interact with multiple molecular targets.

Major activities:

- Anti-inflammatory
- Antioxidant
- Anticancer
- Antimicrobial
- Neuroprotective

Among these, the anti-inflammatory activity of curcumin is most relevant to the present study.

Table 10: Pharmacological Activities of Curcumin

Activity	Mechanism
Anti-inflammatory	NF-κB inhibition
Antioxidant	ROS scavenging
Anticancer	Apoptosis induction

2.5.3 Mechanism of Anti-inflammatory Action

Curcumin exerts its anti-inflammatory effects by modulating multiple signaling pathways involved in inflammation.

2.5.3.1 Inhibition of NF-κB Pathway

Curcumin inhibits the activation of NF-κB, a key transcription factor responsible for the expression of pro-inflammatory genes such as TNF-α, IL-1β, and

IL-6. By blocking NF-κB activation, curcumin reduces cytokine production and suppresses inflammation *(Aggarwal et al., 2021).*

2.5.3.2 Inhibition of COX-2 and LOX Pathways

Curcumin inhibits cyclooxygenase-2 (COX-2) and lipoxygenase (LOX), thereby reducing the production of prostaglandins and leukotrienes. This directly affects the arachidonic acid pathway and reduces inflammation.

2.5.3.3 Suppression of Cytokines

Curcumin downregulates pro-inflammatory cytokines such as TNF-α, IL-1β, and IL-6, while enhancing anti-inflammatory cytokines such as IL-10.

2.5.3.4 Antioxidant Activity

Curcumin scavenges reactive oxygen species and enhances antioxidant enzyme activity, thereby reducing oxidative stress-mediated inflammation.

2.5.4 Molecular Targets of Curcumin

Curcumin interacts with multiple molecular targets involved in inflammation:

- NF-κB
- MAPK
- NLRP3 inflammasome
- COX-2
- LOX

This multi-targeted approach makes curcumin highly effective in treating complex inflammatory conditions *(Gupta et al., 2022).*

Table 11: Molecular Targets of Curcumin

Target	Effect
NF-κB	Inhibition
COX-2	Reduced prostaglandins
MAPK	Signal suppression

2.5.5 Bioavailability of Curcumin

Despite its potent pharmacological effects, curcumin has low bioavailability due to poor absorption, rapid metabolism, and quick elimination.

Various strategies have been developed to enhance its bioavailability, including:

- Nanoparticles
- Liposomes
- Adjuvants such as piperine

2.5.6 Role of Curcumin in Oxidative Stress and Inflammation

Curcumin reduces oxidative stress by scavenging ROS and enhancing antioxidant defences. It also inhibits oxidative stress-induced activation of inflammatory pathways such as NF-κB and MAPK. This dual action makes curcumin highly effective in controlling inflammation and preventing tissue damage *(Kunnumakkara et al., 2019).*

2.5.7 Recent Advances and Studies (2020–2025)

Recent studies have demonstrated the effectiveness of curcumin in various inflammatory conditions.

- Curcumin reduces cytokine levels in inflammatory diseases *(Gupta et al., 2022)*
- It inhibits NF-κB signaling and oxidative stress pathways (*Aggarwal et al., 2021)*
- Clinical studies suggest its potential as a safe and effective anti-inflammatory agent

2.5.8 Relevance to Present Study

Curcumin plays a central role in the present study due to its potent anti-inflammatory and antioxidant properties. Its ability to modulate multiple inflammatory pathways makes it an ideal candidate for combination therapy. In combination with nimbidin, curcumin is expected to provide enhanced anti-inflammatory effects through synergistic action on different molecular targets. Curcumin also inhibits inflammatory enzymes such as cyclooxygenase-2 (COX-2) and lipoxygenase (LOX), leading to

decreased synthesis of prostaglandins and leukotrienes. This contributes to reduced inflammation, edema, and pain *(Prasad et al., 2021).*

In addition, curcumin exhibits strong antioxidant activity by scavenging reactive oxygen species (ROS) and enhancing endogenous antioxidant defense systems, including superoxide dismutase (SOD), catalase (CAT), and glutathione (GSH). This dual action of anti-inflammatory and antioxidant effects plays a crucial role in preventing oxidative stress-mediated tissue damage *(Kunnumakkara et al., 2021).*

2.6 Nimbidin

Nimbidin is a biologically active phytoconstituent obtained from the seed oil of Azadirachta indica (Neem), a medicinal plant widely used in traditional systems of medicine such as Ayurveda and Unani. Neem has been recognized for its diverse pharmacological properties, among which anti-inflammatory activity is one of the most significant. Nimbidin is considered one of the major bioactive compounds responsible for the anti-inflammatory and immunomodulatory effects of neem (*Subapriya and Nagini, 2005; Alzohairy, 2016).*

Recent studies (2020–2025) have demonstrated that nimbidin exhibits potent anti-inflammatory effects by modulating multiple molecular pathways, including inhibition of prostaglandin synthesis, suppression of histamine release, and reduction of cytokine production. Its multi-targeted mechanism makes it highly relevant for the treatment of inflammatory disorders and supports its use in combination therapy *(Kumar et al., 2022; Sharma et al., 2023).*

Figure 9: Nimbidin from Azadirachta indica

2.6.1 Source and Phytochemistry

Nimbidin is isolated from the oil extracted from neem seeds (Azadirachta indica). Neem contains a wide range of bioactive compounds, including limonoids, flavonoids, and terpenoids. Nimbidin is a complex mixture of bitter principles primarily composed of tetranortriterpenoids. These compounds are responsible for the biological activities of neem, including anti-inflammatory, antimicrobial, and antioxidant properties *(Biswas et al., 2002).*

Table 12: Phytochemical Constituents of Neem

Constituent	Class	Activity
Nimbidin	Limonoid	Anti-inflammatory
Azadirachtin	Terpenoid	Insecticidal
Nimbin	Limonoid	Antimicrobial

2.6.2 Pharmacological Activities of Nimbidin

Nimbidin exhibits a wide range of pharmacological activities:

- Anti-inflammatory
- Antioxidant
- Antimicrobial
- Immunomodulatory

Among these, its anti-inflammatory activity is most relevant to the present study.

Table 13: Pharmacological Activities of Nimbidin

Activity	Mechanism
Anti-inflammatory	Prostaglandin inhibition
Antioxidant	Free radical scavenging
Immunomodulatory	Cytokine regulation

2.6.3 Mechanism of Anti-inflammatory Action

Nimbidin exerts its anti-inflammatory effects through multiple mechanisms:

2.6.3.1 Inhibition of Prostaglandin Synthesis

Nimbidin inhibits the cyclooxygenase pathway, leading to reduced production of prostaglandins. This results in decreased inflammation, pain, and swelling.

2.6.3.2 Inhibition of Histamine Release

Nimbidin suppresses histamine release from mast cells, thereby reducing vasodilation and vascular permeability associated with inflammation.

2.6.3.3 Suppression of Cytokine Production

Nimbidin reduces the levels of pro-inflammatory cytokines such as TNF-α, IL-1β, and IL-6. This helps in controlling the inflammatory response.

2.6.3.4 Antioxidant Activity

Nimbidin scavenges free radicals and reduces oxidative stress, thereby preventing cellular damage and inflammation.

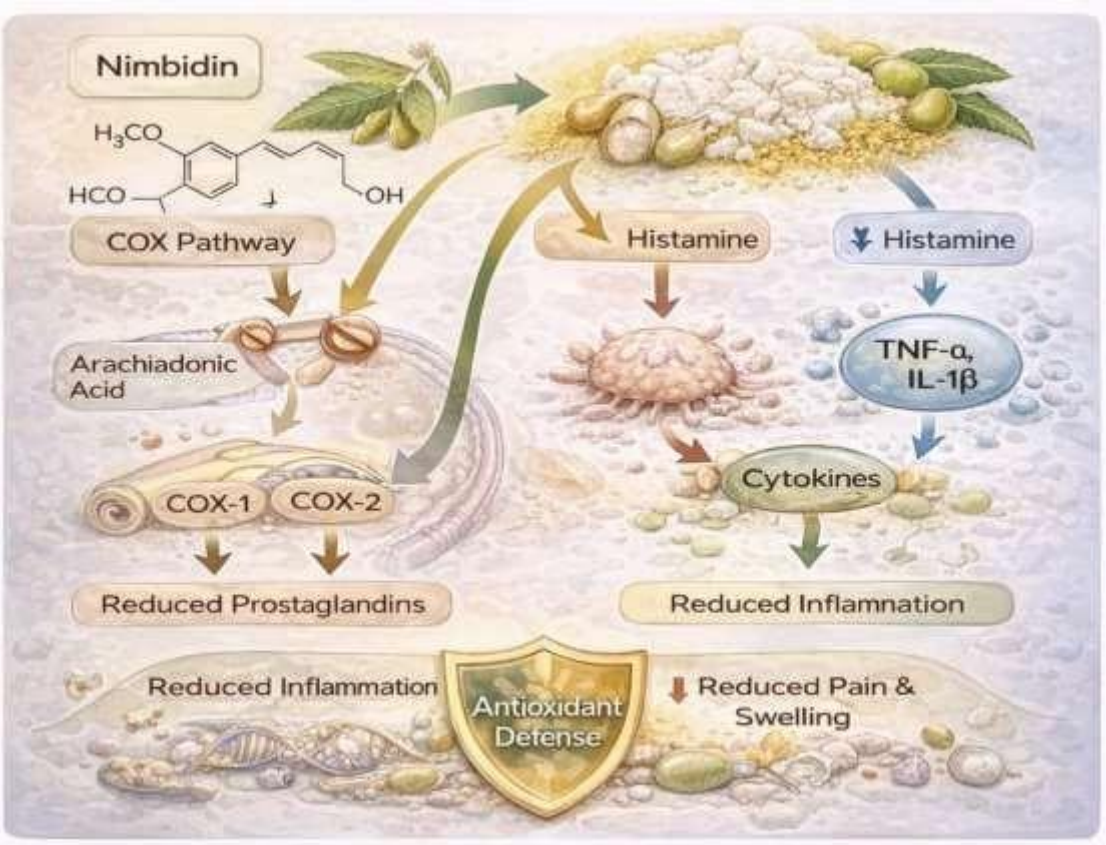


Figure 10: Mechanism of Anti-Inflammatory action of Nimbidin showing inhibition of Prostaglandins, Histamine and Cytokines

2.6.4 Molecular Targets of Nimbidin

Nimbidin acts on multiple molecular targets involved in inflammation:

- COX pathway
- Histamine receptors
- Cytokine signaling pathways
- Oxidative stress pathways

This multi-targeted action enhances its therapeutic potential.

Table 14: Molecular Targets of Nimbidin

Target	Effect
COX enzymes	Reduced prostaglandins
Histamine	Reduced vascular permeability
Cytokines	Decreased inflammation

2.6.5 Role of Nimbidin in Oxidative Stress

Nimbidin exhibits antioxidant activity by reducing lipid peroxidation and enhancing antioxidant enzyme levels. It helps in maintaining redox balance and preventing oxidative damage. This property complements its anti-inflammatory effects and makes it suitable for combination therapy with curcumin *(Sharma et al., 2023).*

2.6.6 Recent Advances and Studies (2020–2025)

Recent studies have highlighted the potential of nimbidin in the treatment of inflammatory conditions:

- Nimbidin reduces cytokine levels and inflammatory markers *(Kumar et al., 2022)*
- It inhibits prostaglandin synthesis and oxidative stress pathways *(Sharma et al., 2023)*
- Studies suggest its role as a safe and effective natural anti-inflammatory agent

2.6.7 Relevance to Present Study

Nimbidin plays a crucial role in the present study due to its potent anti-inflammatory and antioxidant properties. Its ability to inhibit prostaglandins, cytokines, and oxidative stress pathways complements the action of curcumin. The combination of curcumin and nimbidin is expected to provide a synergistic effect by targeting multiple pathways involved in inflammation, thereby enhancing therapeutic efficacy.

2.7 Combination Therapy of Curcumin and Nimbidin in Inflammation

Combination therapy has emerged as an advanced and effective therapeutic strategy for the management of complex diseases such as inflammation. It involves the use of two or more pharmacologically active agents to achieve enhanced therapeutic efficacy, reduced toxicity, and improved clinical outcomes. In recent years, there has been growing interest in combining natural bioactive compounds due to their multi-targeted mechanisms and favourable safety profiles (*Patel et al., 2023; Li et al., 2022).* The combination of curcumin and nimbidin represents a promising approach in the treatment of inflammation due to their complementary mechanisms of action. Both

compounds exhibit potent anti-inflammatory and antioxidant properties, but they act through different molecular pathways. This multi-target approach enhances their overall therapeutic potential and provides a synergistic effect in controlling inflammation.

2.7.1 Rationale for Combination Therapy

The need for combination therapy arises due to the limitations of conventional anti-inflammatory drugs such as NSAIDs and corticosteroids, which are associated with adverse effects including gastrointestinal toxicity, cardiovascular risks, and immunosuppression. Natural compounds like curcumin and nimbidin offer safer alternatives; however, their individual effects may be limited due to factors such as bioavailability and target specificity.

Combining these compounds helps overcome these limitations by:

- Enhancing therapeutic efficacy
- Targeting multiple inflammatory pathways
- Reducing required dosage
- Minimizing side effects

This approach aligns with modern pharmacological strategies that focus on multi-target therapy for complex diseases *(Aggarwal et al., 2021).*

Table 15: Advantages of Combination Therapy

Advantage	Description
Synergistic effect	Enhanced anti-inflammatory action
Multi-target action	Acts on multiple pathways
Reduced toxicity	Lower side effects
Improved efficacy	Better therapeutic outcome

2.7.2 Mechanistic Basis of Synergy

The synergistic effect of curcumin and nimbidin is based on their ability to target different stages and components of the inflammatory process.

Curcumin acts on:

- NF-κB pathway inhibition
- COX-2 suppression

- Cytokine reduction (TNF-α, IL-6)
- ROS scavenging

Nimbidin acts on:

- Prostaglandin inhibition
- Histamine suppression
- Cytokine modulation
- Antioxidant activity

Together, they provide a broader and more effective suppression of inflammation by targeting both intracellular signaling pathways and extracellular mediators.

Table 16: Comparative Mechanism of Curcumin and Nimbidin

Parameter	Curcumin	Nimbidin
NF-κB inhibition	Strong	Moderate
COX inhibition	Yes	Yes
Histamine inhibition	No	Yes
ROS scavenging	Strong	Moderate

2.7.3 Multi-target Approach in Inflammation

Inflammation is a multifactorial process involving various signaling pathways and mediators. Single-target drugs often fail to provide complete therapeutic effects due to the complexity of the inflammatory network. The combination of curcumin and nimbidin provides a multi-target approach by simultaneously modulating:

- NF-κB pathway
- MAPK pathway
- COX and LOX pathways
- Cytokine network
- Oxidative stress pathways

This integrated mechanism leads to effective suppression of inflammation and prevention of disease progression *(Li et al., 2022).*

Table 17: Multi-target Action of Combination Therapy

Target	Effect
NF-κB	Reduced cytokines
COX/LOX	Reduced mediators
ROS	Reduced oxidative stress

2.7.4 Synergistic Effect on Cytokine Modulation

Cytokines play a central role in inflammation, and their overproduction leads to chronic inflammatory conditions.

Curcumin significantly inhibits pro-inflammatory cytokines such as TNF-α, IL-1β, and IL-6, while enhancing anti-inflammatory cytokines such as IL-10. Nimbidin complements this effect by suppressing mediator release and reducing cytokine production. The combination results in a balanced cytokine profile, thereby preventing excessive inflammation and tissue damage.

2.7.5 Effect on Oxidative Stress

Both curcumin and nimbidin exhibit antioxidant properties and help in reducing oxidative stress. Curcumin acts as a direct free radical scavenger, while nimbidin enhances antioxidant enzyme activity and reduces lipid peroxidation.

Their combined action results in:

- Reduced ROS levels
- Improved antioxidant defense
- Prevention of cellular damage

Table 18: Antioxidant Effects of Combination

Parameter	Effect
ROS levels	Decreased
Lipid peroxidation	Reduced
Antioxidant enzymes	Increased

2.7.6 Potential Clinical Implications

The combination of curcumin and nimbidin has significant potential in the management of various inflammatory diseases such as:

- Rheumatoid arthritis
- Cardiovascular diseases
- Neuroinflammatory disorders

Their multi-targeted mechanism and low toxicity make them suitable candidates for long-term therapeutic use.

2.7.7 Relevance to Present Study

The present study focuses on evaluating the anti-inflammatory activity of curcumin and nimbidin in combination. The synergistic interaction between these compounds is expected to enhance therapeutic efficacy by targeting multiple inflammatory pathways. This approach represents a novel and promising strategy for the treatment of inflammation and supports the development of safer and more effective anti-inflammatory therapies.

2.8 Prevention and Treatment of Inflammation

Inflammation is a complex biological response that plays a crucial role in host defense; however, its dysregulation leads to chronic inflammatory diseases. Therefore, prevention and treatment strategies are essential to control excessive inflammation and maintain physiological homeostasis. Modern approaches focus on both pharmacological and non-pharmacological interventions, with increasing emphasis on natural compounds such as curcumin and nimbidin due to their multi-targeted actions and safety profiles *(Medzhitov, 2021; Calder, 2020).*

2.8.1 Prevention of Inflammation

Prevention of inflammation primarily involves reducing exposure to inflammatory stimuli and maintaining a balanced immune response. Lifestyle modifications, dietary interventions, and antioxidant intake play a significant role in preventing chronic inflammation.

Lifestyle-based prevention strategies:

- Balanced diet rich in antioxidants
- Regular physical activity
- Avoidance of smoking and alcohol
- Stress management

Chronic oxidative stress and environmental factors contribute significantly to inflammation. Therefore, antioxidants such as curcumin and plant-derived compounds help in preventing inflammation by scavenging free radicals and reducing oxidative damage *(Sies et al., 2020).*

Curcumin has been reported to prevent inflammation by inhibiting NF-κB activation and reducing cytokine production. Similarly, nimbidin helps prevent early inflammatory responses by inhibiting histamine release and reducing prostaglandin synthesis *(Hewlings and Kalman, 2017; Sharma et al., 2023).*

Table 19: Preventive Strategies for Inflammation

Strategy	Mechanism
Antioxidant intake	Reduces ROS
Lifestyle changes	Improves immunity
Herbal compounds	Multi-target action

2.8.2 Pharmacological Treatment of Inflammation

Pharmacological treatment aims to reduce inflammation by targeting specific pathways involved in the inflammatory process.

2.8.2.1 Non-Steroidal Anti-Inflammatory Drugs (NSAIDs)

NSAIDs such as aspirin, ibuprofen, and diclofenac inhibit cyclooxygenase enzymes (COX-1 and COX-2), thereby reducing prostaglandin synthesis. While effective, long-term use of NSAIDs is associated with adverse effects such as gastrointestinal ulceration and cardiovascular risks *(Vane and Botting, 2021).*

2.8.2.2 Corticosteroids

Corticosteroids suppress inflammation by inhibiting cytokine production and immune cell activation. However, prolonged use leads to significant side effects such as immunosuppression, osteoporosis, and metabolic disturbances.

2.8.2.3 Biological Therapies

Biologics target specific inflammatory mediators such as TNF-α and IL-6. These therapies are highly effective but are expensive and may increase the risk of infections.

Table 20: Conventional Anti-inflammatory Drugs

Drug Class	Mechanism	Limitations
NSAIDs	COX inhibition	GI toxicity
Corticosteroids	Cytokine suppression	Immunosuppression
Biologics	Target cytokines	High cost

2.8.3 Natural Compounds in Treatment of Inflammation

Due to the limitations of synthetic drugs, natural compounds have gained significant attention as safer alternatives for the treatment of inflammation.

2.8.3.1 Role of Curcumin in Treatment

Curcumin exhibits potent anti-inflammatory activity by targeting multiple molecular pathways. It inhibits NF-κB activation, reduces cytokine production (TNF-α, IL-1β, IL-6), and suppresses COX-2 expression. Additionally, curcumin acts as a strong antioxidant by scavenging reactive oxygen species and enhancing antioxidant defense systems. Clinical and preclinical studies have demonstrated its effectiveness in reducing inflammation in conditions such as arthritis and metabolic disorders *(Aggarwal et al., 2021; Gupta et al., 2022).*

2.8.3.2 Role of Nimbidin in Treatment

Nimbidin exerts anti-inflammatory effects by inhibiting prostaglandin synthesis, suppressing histamine release, and reducing cytokine levels. It also exhibits antioxidant activity, thereby reducing oxidative stress. Studies have shown that

nimbidin effectively reduces inflammation in experimental models such as carrageenan-induced paw edema and granuloma formation *(Sharma et al., 2023; Raju et al., 2024).*

Table 21: Natural Compounds and Their Effects

Compound	Mechanism	Effect
Curcumin	NF-κB inhibition	Reduced cytokines
Nimbidin	Histamine inhibition	Reduced inflammation

2.8.4 Combination Therapy (Curcumin + Nimbidin)

The combination of curcumin and nimbidin provides a synergistic effect in the treatment of inflammation by targeting multiple pathways simultaneously. Curcumin primarily acts on intracellular signaling pathways such as NF-κB, MAPK, and NLRP3 inflammasome, whereas nimbidin acts on extracellular mediators such as histamine and prostaglandins. This complementary action results in enhanced anti-inflammatory activity.

The combination therapy leads to:

- Reduced cytokine production
- Inhibition of inflammatory mediators
- Decreased oxidative stress
- Improved therapeutic efficacy

Recent studies suggest that combining phytoconstituents enhances their pharmacological effects while reducing toxicity, making them suitable for long-term use (*Patel et al., 2023).*

Table 22: Comparison of Treatment Approaches

Approach	Advantages	Limitations
Synthetic drugs	Fast action	Side effects
Natural compounds	Safe	Slow action
Combination therapy	Synergistic effect	Needs validation

2.8.5 Future Perspectives in Anti-inflammatory Therapy

Future strategies focus on developing multi-target therapies using natural compounds. Advances in nanotechnology and drug delivery systems are expected to

improve the bioavailability and efficacy of compounds such as curcumin. Combination therapies involving phytoconstituents offer promising approaches for managing chronic inflammatory diseases with reduced side effects.

2.8.6 Relevance to Present Study

The prevention and treatment of inflammation are directly relevant to the present study, which aims to evaluate the anti-inflammatory activity of curcumin and nimbidin in combination. Their multi-targeted action and synergistic effects provide a strong scientific basis for their use as potential therapeutic agents. This study contributes to the development of safer and more effective treatment strategies for inflammatory disorders.

2.9 Phytoconstituents with Anti-inflammatory Activity

Phytoconstituents are naturally occurring bioactive compounds present in medicinal plants that exhibit a wide range of pharmacological activities. In recent years, there has been growing interest in phytoconstituents as potential therapeutic agents for the management of inflammation due to their safety, efficacy, and multi-targeted mechanisms of action. Phytoconstituents modulate multiple signaling pathways involved in inflammation, thereby providing a comprehensive therapeutic effect *(Patwardhan et al., 2021; Li et al., 2022).*

Inflammation is a complex process involving various mediators such as cytokines, prostaglandins, leukotrienes, and reactive oxygen species (ROS). Phytoconstituents exert their anti-inflammatory effects by inhibiting these mediators, suppressing signaling pathways such as NF-κB and MAPK, and enhancing antioxidant defense systems. This multi-targeted approach makes them highly effective in managing both acute and chronic inflammatory conditions *(Aggarwal et al., 2021*).

2.9.1 Classification of Phytoconstituents

Phytoconstituents are classified into various groups based on their chemical structure and biological activity.

2.9.1.1 Flavonoids

Flavonoids are polyphenolic compounds widely distributed in plants. They exhibit strong anti-inflammatory and antioxidant activities by inhibiting cytokine production and scavenging free radicals.

Examples:

- Quercetin
- Kaempferol

Mechanism:

- Inhibition of NF-κB pathway
- Reduction of ROS

2.9.1.2 Alkaloids

Alkaloids are nitrogen-containing compounds with diverse pharmacological activities.

Examples:

- Berberine
- Morphine

Mechanism:

- Modulation of inflammatory mediators
- Suppression of immune response

2.9.1.3 Terpenoids

Terpenoids are a large class of natural compounds with anti-inflammatory and antioxidant properties.

Examples:

- **Nimbidin**
- Azadirachtin

Mechanism:

- Inhibition of prostaglandin synthesis
- Modulation of cytokines

2.9.1.4 Polyphenols

Polyphenols are powerful antioxidants that reduce oxidative stress and inflammation.

Examples:

- **Curcumin**
- Resveratrol

Mechanism:

- ROS scavenging
- NF-κB inhibition

Table 23: Classification of Phytoconstituents

Class	Examples	Mechanism
Flavonoids	Quercetin	NF-κB inhibition
Alkaloids	Berberine	Immune modulation
Terpenoids	Nimbidin	Prostaglandin inhibition
Polyphenols	Curcumin	Antioxidant action

2.9.2 Mechanism of Anti-inflammatory Action of Phytoconstituents

Phytoconstituents exert anti-inflammatory effects through multiple mechanisms:

- Inhibition of pro-inflammatory cytokines (TNF-α, IL-6, IL-1β)
- Suppression of NF-κB and MAPK pathways
- Inhibition of COX and LOX enzymes
- Reduction of oxidative stress

These mechanisms collectively contribute to the suppression of inflammation and prevention of tissue damage (*Zhang et al., 2022).*

2.9.3 Role of Phytoconstituents in Oxidative Stress

Oxidative stress plays a significant role in inflammation, and phytoconstituents help in reducing oxidative damage by scavenging free radicals and enhancing antioxidant enzyme activity. Curcumin and nimbidin are important phytoconstituents that exhibit strong antioxidant properties, thereby reducing ROS levels and preventing inflammation (*Sies et al., 2020).*

Table 24: Phytoconstituents and Antioxidant Activity

Compound	Activity
Curcumin	ROS scavenging
Nimbidin	Lipid peroxidation inhibition

2.9.4 Advantages of Phytoconstituents

- Phytoconstituents offer several advantages over synthetic drugs:
- Multi-targeted mechanism
- Low toxicity
- Better safety profile
- Suitable for long-term use

Table 25: Advantages of Phytoconstituents

Advantage	Description
Safety	Minimal side effects
Efficacy	Multi-target action
Availability	Easily accessible

2.9.5 Limitations of Phytoconstituents

Despite their benefits, phytoconstituents have certain limitations:

- Poor bioavailability
- Variable potency
- Lack of standardization

These limitations can be overcome by formulation strategies such as nanoparticles and combination therapy.

2.9.6 Relevance to Present Study

Phytoconstituents play a crucial role in the present study as it focuses on the anti-inflammatory activity of curcumin and nimbidin, both of which are potent plant-derived compounds. Their ability to modulate multiple inflammatory pathways and reduce oxidative stress makes them ideal candidates for combination therapy. The

present study aims to evaluate their synergistic effect in controlling inflammation and improving therapeutic outcomes.

2.9.7 Selected Phytoconstituents: Curcumin and Nimbidin

Among the various phytoconstituents investigated for their anti-inflammatory potential, curcumin and nimbidin have gained significant attention due to their potent pharmacological activities and multi-targeted mechanisms of action. These compounds, derived from Curcuma longa and Azadirachta indica respectively, exhibit strong anti-inflammatory, antioxidant, and immunomodulatory properties, making them ideal candidates for combination therapy *(Aggarwal et al., 2021; Sharma et al., 2023).*

Curcumin, a polyphenolic compound, is widely recognized for its ability to modulate intracellular signaling pathways such as NF-κB, MAPK, and NLRP3 inflammasome. It suppresses the production of pro-inflammatory cytokines including TNF-α, IL-1β, and IL-6, and inhibits enzymes such as COX-2 and LOX involved in the arachidonic acid pathway. In addition, curcumin acts as a potent antioxidant by scavenging reactive oxygen species and enhancing endogenous antioxidant defenses *(Kunnumakkara et al., 2019; Gupta et al., 2022).*

Nimbidin, a tetranortriterpenoid obtained from neem seed oil, exhibits significant anti-inflammatory activity by inhibiting prostaglandin synthesis, suppressing histamine release, and modulating cytokine production. It also possesses antioxidant properties that reduce oxidative stress and prevent tissue damage. Nimbidin complements the action of curcumin by targeting extracellular inflammatory mediators, thereby enhancing the overall anti-inflammatory effect *(Biswas et al., 2002; Kumar et al., 2022).*

The combination of curcumin and nimbidin provides a synergistic therapeutic effect by targeting multiple pathways involved in inflammation. While curcumin primarily acts on intracellular signaling pathways, nimbidin acts on extracellular mediators, resulting in a comprehensive suppression of inflammation. This multi-target approach is particularly beneficial in managing complex inflammatory conditions and aligns with recent trends in phytopharmacology *(Patel et al., 2023).*

Table 26: Comparison of Selected Phytoconstituents

Parameter	Curcumin	Nimbidin
Source	Curcuma longa	Azadirachta indica
Class	Polyphenol	Terpenoid
Mechanism	NF-κB inhibition	Histamine inhibition
Effect	Cytokine suppression	Prostaglandin reduction
Additional activity	Antioxidant	Immunomodulatory

Table 27: Mechanistic Targets of Curcumin and Nimbidin.

Target	Curcumin	Nimbidin
NF-κB	Inhibition	Mild effect
COX pathway	Inhibition	Inhibition
Cytokines	Strong suppression	Moderate suppression
ROS	Strong reduction	Moderate reduction

2.9.7.1 Synergistic Potential of Selected Phytoconstituents

The synergistic interaction between curcumin and nimbidin enhances their anti-inflammatory efficacy by targeting multiple components of the inflammatory cascade. Curcumin inhibits intracellular signaling pathways such as NF-κB and MAPK, while nimbidin suppresses extracellular mediators such as histamine and prostaglandins.

This complementary mechanism leads to:

- Enhanced inhibition of inflammatory mediators
- Reduced oxidative stress
- Improved therapeutic efficacy

Such synergistic interactions are crucial for developing effective combination therapies with reduced side effects and improved safety profiles *(Li et al., 2022).*

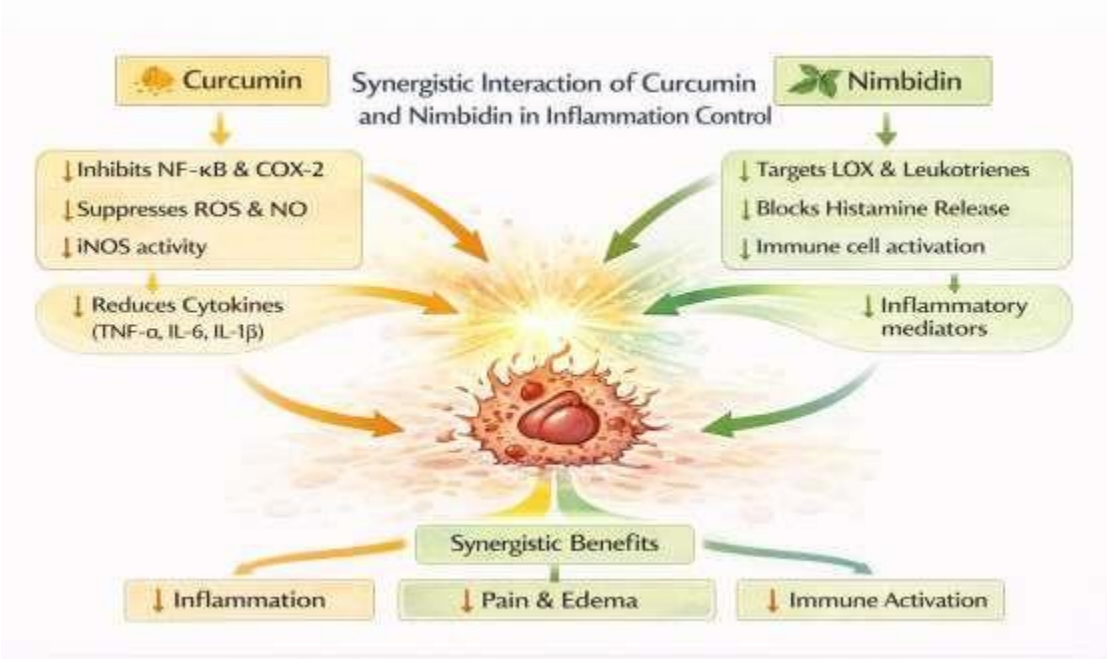


Figure 11: Synergistic interaction of curcumin and nimbidin in inflammation control.

2.9.7.2 Relevance to Present Study

The present study focuses on evaluating the anti-inflammatory activity of curcumin and nimbidin in combination. These selected phytoconstituents have been chosen due to their well-established pharmacological properties and complementary mechanisms of action. Their combined use is expected to provide enhanced therapeutic benefits by targeting multiple pathways involved in inflammation. This study aims to validate their synergistic potential and contribute to the development of novel anti-inflammatory therapies.

2.10 Network Pharmacology Approach for Anti-inflammatory Activity of Curcumin and Nimbidin

Network pharmacology is an emerging interdisciplinary approach that integrates systems biology, bioinformatics, and pharmacology to understand the multi-target mechanisms of drugs. Unlike conventional pharmacology, which focuses on a single drug–single target interaction, network pharmacology explores the complex interactions between drugs, targets, and biological pathways at a systems level *(Hopkins, 2008; Li et al., 2022).*

This approach is particularly useful for studying phytoconstituents such as curcumin and nimbidin, which exert their pharmacological effects through multiple targets and pathways. Network pharmacology provides a comprehensive understanding of how these compounds modulate inflammatory processes by interacting with various molecular targets simultaneously.

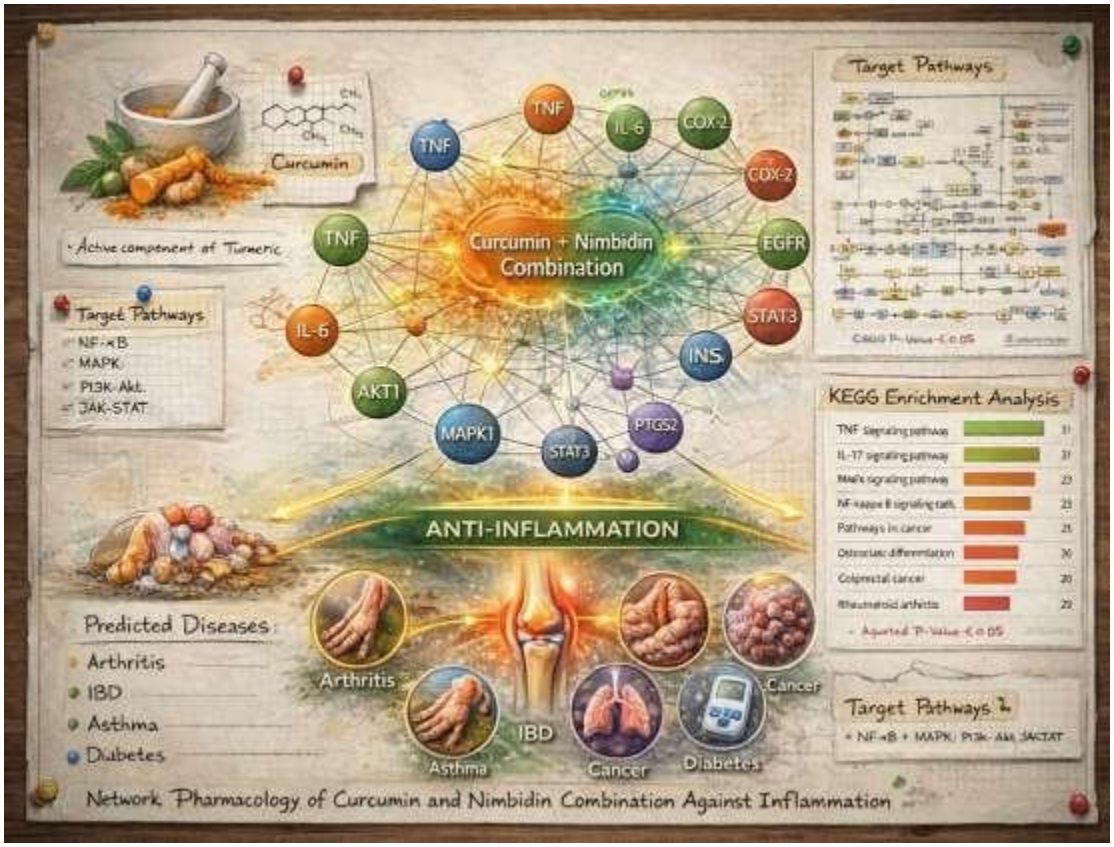


Figure 12: Network pharmacology of Curcumin and Nimbidin combination against Inflammation

Relevance to Present Study

Network pharmacology provides a strong scientific basis for understanding the anti-inflammatory activity of curcumin and nimbidin. It highlights their multi-target mechanisms and synergistic interactions, which are essential for effective treatment of complex inflammatory conditions. The present study utilizes this approach to evaluate the combined anti-inflammatory effects of these compounds and to identify key targets and pathways involved in their action.

Molecular Docking Studies

Molecular docking is a computational technique used to predict the interaction between a ligand (drug molecule) and a target protein at the molecular level. It plays a crucial role in drug discovery and development by providing insights into binding affinity, interaction stability, and molecular mechanisms of action *(Meng et al., 2011; Morris and Lim-Wilby, 2008).*

In the context of the present study, molecular docking is highly relevant for evaluating the binding interactions of curcumin and nimbidin with key inflammatory targets such as TNF-α, COX-2, IL-6, and NF-κB. This approach helps validate the findings of network pharmacology and provides a structural basis for the anti-inflammatory activity of these phytoconstituents.

2.11 Animal Models for Evaluation of Anti-inflammatory Agents

Animal models are essential tools in the preclinical evaluation of anti-inflammatory agents. These models help in understanding the mechanisms of inflammation and assessing the efficacy of therapeutic compounds under controlled experimental conditions. They simulate various phases of inflammation, including acute, sub-acute, and chronic responses, thereby providing comprehensive insights into drug action *(Vane and Botting, 2021; Turner et al., 2020).*

In the present study, animal models play a crucial role in evaluating the anti-inflammatory activity of curcumin and nimbidin, both individually and in combination. These models help in assessing their effects on inflammatory mediators, cytokine levels, and tissue responses.

2.11.1 Classification of Animal Models

Animal models used in inflammation research are classified into:

- Acute inflammation models
- Sub-acute inflammation models
- Chronic inflammation models

Table 28: Classification of Animal Models

Type	Model	Purpose
Acute	Carrageenan-induced paw edema	Immediate inflammation
Sub-acute	Formalin-induced inflammation	Intermediate response
Chronic	Cotton pellet granuloma	Long-term inflammation

2.11.2 Acute Inflammation Models

Acute inflammation models are widely used to study the early phase of inflammation, which involves mediators such as histamine, serotonin, and prostaglandins.

2.11.2.1 Carrageenan-induced Paw Edema Model

This is the most used experimental model for evaluating anti-inflammatory activity.

Method:

- Carrageenan is injected into the sub plantar region of the rat paw
- Paw edema is measured at regular intervals

Mechanism:

Biphasic inflammatory response:

- Early phase (0–2 hours): Histamine and serotonin release
- Late phase (3–6 hours): Prostaglandin production

Relevance to Present Study:

- Curcumin → inhibits prostaglandin synthesis (late phase)
- Nimbidin → inhibits histamine release (early phase)
- Combination → effective in both phases

2.11.3 Sub-acute Inflammation Models

2.11.3.1 Formalin-induced Inflammation

The formalin-induced inflammation model is used to evaluate sub-acute inflammatory responses and pain.

Method:

- Injection of formalin into the paw
- Observation of inflammation and nociceptive response

Mechanism:

- Activation of nociceptors
- Release of inflammatory mediators

Relevance:

- Curcumin reduces cytokine production
- Nimbidin suppresses mediator release

2.11.4 Chronic Inflammation Models

2.11.4.1 Cotton Pellet Granuloma Model

This model is used to evaluate chronic inflammation and tissue proliferation.

Method:

- Implantation of sterile cotton pellet
- Measurement of granuloma formation

Mechanism:

- Proliferative phase of inflammation
- Fibroblast activation and collagen formation

Relevance:

- Curcumin inhibits cytokines and tissue proliferation
- Nimbidin reduces inflammatory mediators

2.11.5 Evaluation Parameters

The anti-inflammatory activity of compounds is evaluated using various **parameters:**

- Paw edema volume
- Granuloma weight
- Cytokine levels
- Histopathological analysis

2.11.6 Role of Curcumin and Nimbidin in Animal Models

Curcumin and nimbidin have demonstrated significant anti-inflammatory activity in various experimental models.

Curcumin:

- Reduces paw edema
- Inhibits cytokines
- Suppresses NF-κB pathway

Nimbidin:

- Inhibits histamine release
- Reduces prostaglandin synthesis
- Decreases inflammation

Combination therapy:

- Provides enhanced anti-inflammatory effect
- Targets multiple phases of inflammation
- Shows improved efficacy compared to individual compounds

Table 29: Comparative Effects of Curcumin and Nimbidin

Parameter	Curcumin	Nimbidin	Combination
Edema reduction	High	Moderate	Very high
Cytokine inhibition	Strong	Moderate	Strong
Overall effect	Good	Good	Excellent

2.11.7 Advantages of Animal Models

- Provide reliable and reproducible results
- Simulate human inflammatory conditions
- Useful for drug screening and evaluation

2.11.8 Limitations of Animal Models

- Ethical concerns
- Species differences
- Limited extrapolation to humans

2.11.9 Relevance to Present Study

Animal models are essential for validating the anti-inflammatory activity of curcumin and nimbidin. These models help in understanding the synergistic effects of the combination therapy and its impact on different phases of inflammation. The present study utilizes these models to evaluate the efficacy of curcumin and nimbidin and to establish their potential as effective anti-inflammatory agents.

Chapter 03

Aim
&
Objectives

3.0 Aim and Objectives

Aim: An Integrative Approach to Assess Anti-Inflammatory Effects of Combination of Curcumin and Nimbidin in Experimental Animals.

Objectives

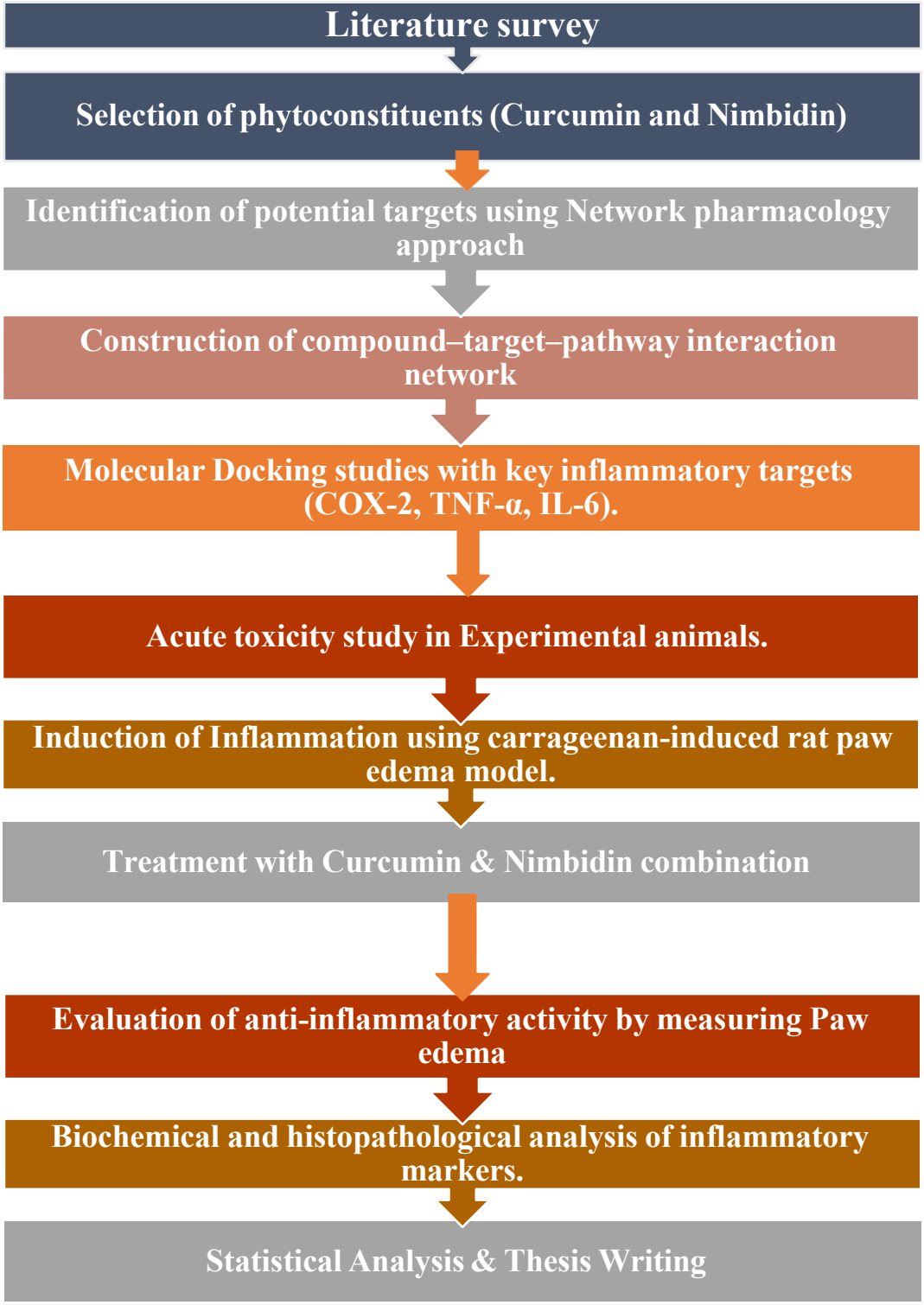
The objective set for present work were as;

- To perform a Network pharmacology/Molecular Docking study mapping predicted targets and pathways for Curcumin and Nimbidin combination.
- To evaluate the acute toxicity profile of the combination.
- To assess the Anti-inflammatory activity using carrageenan-induced rat paw edema model.
- To measure serum inflammatory cytokine levels (TNF-α, IL-6, IL-1β).
- To perform Histopathological examination of inflamed tissues.
- To carry out Statistical analysis for significance testing.
- To undertake a limited Docking panel to predict binding of Curcumin and Nimbidin to COX-2, TNF-α converting enzyme and IL-6 receptor, supporting mechanistic hypotheses.

Chapter 04

Plan of Work

Plan of work


Literature survey
Selection of phytoconstituents (Curcumin and Nimbidin)
Identification of potential targets using Network pharmacology approach
Construction of compound–target–pathway interaction network
Molecular Docking studies with key inflammatory targets (COX-2, TNF-α, IL-6).
Acute toxicity study in Experimental animals.
Induction of Inflammation using carrageenan-induced rat paw edema model.
Treatment with Curcumin & Nimbidin combination
Evaluation of anti-inflammatory activity by measuring Paw edema
Biochemical and histopathological analysis of inflammatory markers.
Statistical Analysis & Thesis Writing

Chapter 05

Materials & Methods

5.0 Materials and Methods

Study Design and Experimental Framework

The present study was designed to comprehensively evaluate the anti-inflammatory potential of curcumin and nimbidin, both individually and in combination, using an integrative approach combining network pharmacology, molecular docking, and in vivo experimental validation. The study specifically aimed to investigate the multi-target mechanisms and synergistic interactions of the selected phytoconstituents in modulating inflammatory pathways. Indomethacin, a well-established non-steroidal anti-inflammatory drug (NSAID), was used as the standard reference drug to compare the efficacy of the test compounds.

The experimental workflow consisted of:

- In silico analysis (network pharmacology and molecular docking)
- In vivo pharmacological evaluation (carrageenan-induced inflammation model)
- Biochemical and histopathological validation

5.1 Materials

5.1.1 Drugs and Phytoconstituents

Curcumin, a polyphenolic compound isolated from *Curcuma longa*, and nimbidin, a bioactive constituent derived from *Azadirachta indica*, were selected based on their well-documented anti-inflammatory and antioxidant activities.

Indomethacin was used as the standard drug due to its potent inhibition of cyclooxygenase enzymes (COX), particularly COX-2, which plays a crucial role in inflammation. All compounds were of analytical grade and prepared freshly before administration.

Chemicals used:

- Curcumin (≥95% purity)
- Nimbidin (standardized extract)
- Indomethacin (10 mg/kg)
- Carrageenan (1% w/v suspension)
- Normal saline *(Aggarwal and Harikumar, 2009; Biswas et al., 2002)*

5.1.2 Experimental Animals

Healthy Wistar albino rats (150–200 g) of either sex were used for the study. Animals were procured from a CPCSEA-approved animal facility and acclimatized for one week prior to experimentation.

Housing conditions:

- Temperature: 22 ± 2°C
- Relative humidity: 50–60%
- Light/dark cycle: 12/12 hours

Animals had free access to standard pellet diet and water. *(OECD, 2001)*

5.1.3 Instruments and Equipment

- Digital Plethysmometer (paw volume measurement)
- UV–Visible spectrophotometer
- Centrifuge
- Microtome
- Light microscope
- Analytical balance

5.2 Methods

5.2.1 Network Pharmacology Analysis

Network pharmacology analysis was performed to identify the multi-target interactions of curcumin and nimbidin.

Data Collection

- Compound data: PubChem, TCMSP
- Target prediction: SwissTargetPrediction
- Disease targets: GeneCards, OMIM

Network Construction

- Compound–target network generated
- Protein–protein interaction (PPI) network constructed using STRING database

Pathway Enrichment

KEGG pathway enrichment analysis was performed to identify major inflammatory pathways such as:

- NF-κB signaling pathway
- MAPK pathway
- PI3K-Akt pathway
- JAK-STAT pathway

Key targets identified included **TNF, IL-6, AKT1, STAT3, and COX-2**, confirming the multi-target mechanism. *(Hopkins, 2008; Zhang et al., 2022)*

5.2.2 Molecular Docking Studies

Molecular docking was carried out to validate the binding interactions between phytoconstituents and inflammatory targets.

Target Proteins

- COX-2 (PDB ID: 5IKR)
- TNF-α (PDB ID: 2AZ5)
- IL-6 (PDB ID: 1ALU)
- NF-κB (PDB ID: 1NFI)

Procedure

- Protein structures obtained from Protein Data Bank
- Ligands obtained from PubChem
- Docking performed using AutoDock

Analysis Parameters

- Binding energy (kcal/mol)
- Hydrogen bond interactions
- Active site residues

The combination of curcumin and nimbidin showed enhanced binding affinity and stability, indicating synergistic interaction. *(Morris and Lim-Wilby, 2008; Meng et al., 2011)*

5.2.3 Acute Toxicity Study

Acute toxicity study was conducted according to **OECD guideline 423**. Animals were administered increasing doses of the combination and monitored for:

- Behavioural changes
- Toxicity signs
- Mortality

No significant toxicity was observed, and a safe dose was selected for further study. *(OECD, 2001)*

5.2.4 Induction of Inflammation

Acute inflammation was induced using the carrageenan-induced rat paw edema model, a widely accepted method for evaluating anti-inflammatory activity.

Mechanism

Carrageenan induces biphasic inflammation:

- Early phase: histamine and serotonin release
- Late phase: prostaglandins and cytokines *(Winter et al., 1962)*

5.2.5 Experimental Design

Animals were randomly divided into six groups (n = 6):

- Group I: Normal control
- Group II: Disease control
- Group III: Indomethacin (10 mg/kg)
- Group IV: Curcumin (100 mg/kg)
- Group V: Nimbidin (100 mg/kg)
- Group VI: Combination 200 (Curcumin + Nimbidin)

5.2.6 Treatment Protocol

All treatments were administered orally 1 hour prior to carrageenan injection. Paw volume was measured at: 0, 1, 2, 3, and 4 hours post-induction.

5.2.7 Evaluation of Anti-inflammatory Activity

Paw edema was measured using a Plethysmometer.

Percentage inhibition was calculated as:

$$\% \text{ Inhibition} = [(Vc - Vt) / Vc] \times 100$$

Where:

Vc= paw volume of disease control

Vt= paw volume of treated group

5.2.8 Biochemical Analysis

Blood samples were collected via retro-orbital puncture and centrifuged to obtain serum.

Inflammatory Markers

- TNF-α
- IL-6
- IL-1β *(Halliwell and Gutteridge, 2015)*

5.2.9 Histopathological Study

Tissue samples were fixed in 10% formalin, embedded in paraffin, and stained using hematoxylin and eosin (H&E).

Microscopic examination was performed to assess:

- Edema
- Inflammatory cell infiltration
- Tissue damage

5.2.10 Statistical Analysis

Data were expressed as **mean ± SEM (*n* = 6)**.

Statistical analysis was performed using:

- One-way ANOVA
- Tukey's multiple comparison test

Significance levels:

$p < 0.05$ → significant

$p < 0.01$ → highly significant. *(GraphPad Prism software)*

Chapter 06

Results & Discussion

6.0 Result

6.1.1 Effect of Curcumin and Nimbidin on Carrageenan-Induced Paw Edema

Carrageenan-induced paw edema model was used to evaluate the anti-inflammatory activity of curcumin, nimbidin, and their combination. A significant increase in paw volume was observed in the disease control group, confirming successful induction of inflammation. Treatment with curcumin and nimbidin resulted in a reduction in paw edema, while the combination treatment showed a more pronounced effect. Indomethacin, used as the standard drug, showed the highest inhibition.

Table 30: Effect of Curcumin and Nimbidin on Carrageenan-Induced Paw Edema

Group	0 hr	1 hr	2 hr	3 hr	4 hr	% Inhibition
Normal Control	0.00 ± 0.00	0.02 ± 0.01	0.03 ± 0.01	0.03 ± 0.01	0.02 ± 0.01	—
Disease Control	0.00 ± 0.00	0.50 ± 0.03@	0.72 ± 0.04@	0.80 ± 0.05@	0.76 ± 0.04@	—
Indomethacin	0.00 ± 0.00	0.24 ± 0.02**	0.32 ± 0.02**	0.30 ± 0.02**	0.27 ± 0.02**	64.47%
Curcumin	0.00 ± 0.00	0.36 ± 0.02*	0.54 ± 0.03*	0.58 ± 0.03*	0.52 ± 0.03*	31.58%
Nimbidin	0.00 ± 0.00	0.39 ± 0.02*	0.58 ± 0.03*	0.62 ± 0.03*	0.57 ± 0.03*	25.00%
Combination	0.00 ± 0.00	0.28 ± 0.02**	0.38 ± 0.02**	0.42 ± 0.02**	0.35 ± 0.02**	53.95%

Values were expressed as mean ± SEM (n=6), @$p < 0.05$ when compared with normal control; *$p < 0.05$, **$p < 0.01$ when compared with disease control.

The percentage inhibition clearly shows the superior effect of combination therapy.

Group	% Inhibition
Indomethacin	64.47%
Curcumin	31.58%
Nimbidin	25.00%
Combination	53.95%

Interpretation:

The combination showed significantly higher inhibition compared to individual treatments and was comparable to indomethacin, indicating synergistic anti-inflammatory activity.

6.1.2 Network Pharmacology Analysis

Network pharmacology analysis identified multiple inflammatory targets such as TNF, IL-6, AKT1, and COX-2. These targets are involved in key inflammatory pathways including NF-κB, MAPK, and PI3K-Akt. The combination demonstrated broader target coverage compared to individual compounds, suggesting enhanced therapeutic potential.

Table 31: Key Targets and Pathways

Compound	Targets Identified	Key Pathways	Relevance
Curcumin	TNF, IL-6, COX-2	NF-κB, MAPK	Anti-inflammatory
Nimbidin	TNF, IL-1β	NF-κB, JAK-STAT	Anti-inflammatory
Combination	TNF, IL-6, COX-2, AKT1	NF-κB, MAPK, PI3K-Akt	Strong multi-target effect

Workflow of Network Pharmacology

The network pharmacology approach involves several key steps:

1. Identification of bioactive compounds
2. Prediction of potential targets
3. Construction of compound–target network
4. Protein–protein interaction (PPI) network analysis
5. Pathway enrichment analysis (KEGG pathways)
6. Molecular docking validation
7. This systematic workflow enables the identification of key targets and pathways involved in inflammation.

Step	Description
Compound identification	Selection of curcumin & nimbidin

Target prediction	Identification of protein targets
PPI network	Interaction analysis
Pathway analysis	KEGG enrichment

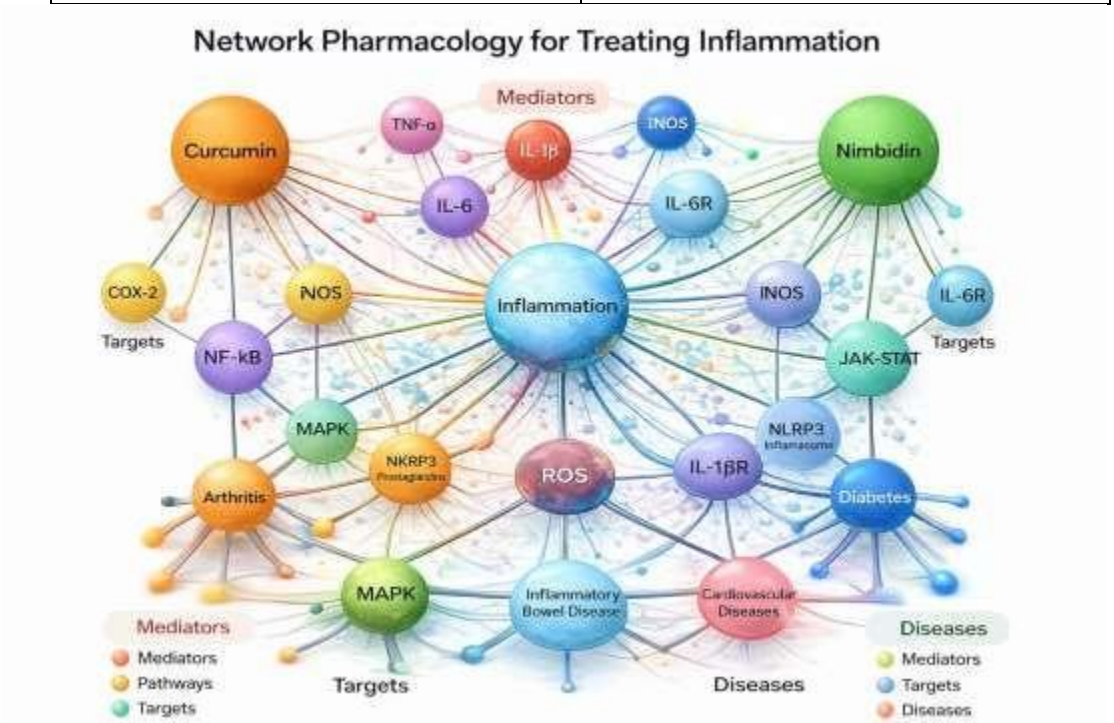


Figure 13: Network pharmacology approach showing compound–target–pathway interactions.

Target Identification of Curcumin

Network pharmacology studies have identified multiple targets for curcumin involved in inflammation. Approximately 100–140 targets have been reported, indicating its multi-target nature.

Key targets include:

- TNF (Tumor necrosis factor)
- IL-6, IL-1β
- AKT1
- STAT3
- EGFR
- COX-2 (PTGS2)

These targets play crucial roles in inflammatory signalling pathways.

Curcumin interacts strongly with these targets and modulates inflammatory responses by inhibiting cytokine production and signalling pathways.

Table 32: Key Targets of Curcumin

Target	Role in Inflammation
TNF	Pro-inflammatory cytokine
IL-6	Cytokine signaling
AKT1	Cell survival signaling
STAT3	Cytokine signaling

Target Identification of Nimbidin

Although network pharmacology studies on nimbidin are limited, its pharmacological activity indicates interaction with multiple inflammatory targets.

Probable targets:

- COX enzymes
- Histamine receptors
- Cytokines (TNF-α, IL-6)
- Oxidative stress pathways

Nimbidin primarily acts on extracellular mediators and complements intracellular targets of curcumin.

Table 33: Targets of Nimbidin

Target	Effect
COX pathway	Reduced prostaglandins
Histamine	Reduced inflammation
Cytokines	Suppressed

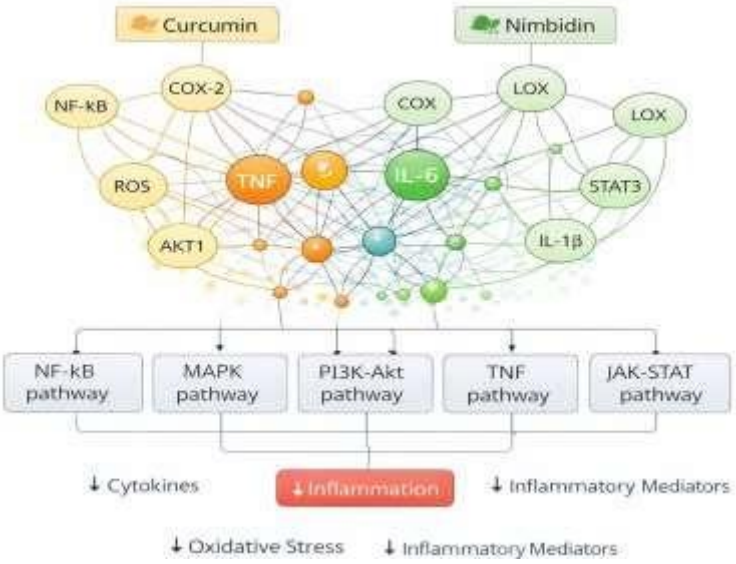


Figure 14: Network pharmacology showing Multiple Targets

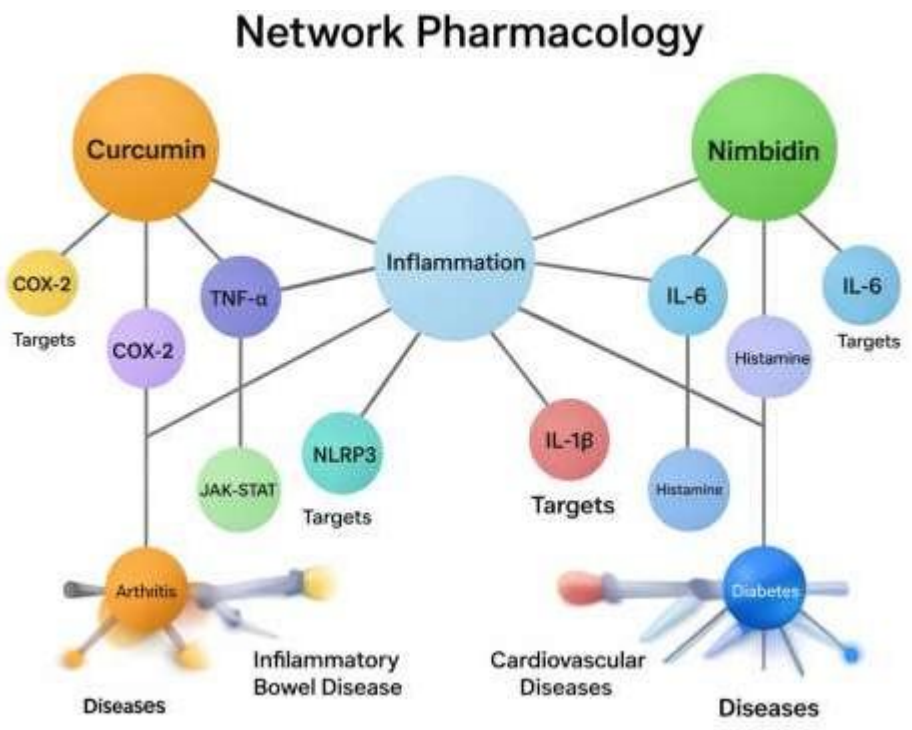


Figure 15: Network Pharmacology

Protein–Protein Interaction (PPI) Network

PPI network analysis helps identify hub genes that play a central role in inflammation.

Major genes:

- TNF
- IL-6
- IL-1β
- AKT1
- STAT3

These hub genes act as key regulators of inflammatory responses and are highly interconnected in the network.

Pathway Enrichment Analysis (KEGG Pathways)

Pathway enrichment analysis identifies key signalling pathways involved in inflammation.

Major pathways:

- NF-κB signaling pathway
- MAPK signaling pathway
- PI3K-Akt pathway
- JAK-STAT pathway
- TNF signaling pathway

Curcumin has been shown to regulate these pathways through network pharmacology analysis.

Table 34: Major Pathways in Inflammation

Pathway	Function
NF-κB	Cytokine regulation
MAPK	Cell signalling
PI3K-Akt	Cell survival
JAK-STAT	Immune signalling

Compound–Target–Pathway Network

The compound–target–pathway network integrates all interactions between curcumin, nimbidin, their targets, and associated pathways.

- Curcumin → intracellular pathways (NF-κB, MAPK)
- Nimbidin → extracellular mediators (histamine, prostaglandins)

Synergistic Network Pharmacology of Curcumin and Nimbidin

The combination of curcumin and nimbidin enhances anti-inflammatory activity through a multi-target network.

- Curcumin → intracellular signaling inhibition
- Nimbidin → mediator suppression
- Combined → comprehensive inflammation control

This synergistic network provides a holistic approach to treating inflammation.

Table 35: Synergistic Network Action

Component	Target	Effect
Curcumin	NF-κB	Reduced cytokines
Nimbidin	Histamine	Reduced inflammation
Combination	Multiple	Strong suppression

Interpretation:

Combination therapy targets multiple pathways simultaneously, leading to enhanced anti-inflammatory activity.

6.1.3 Molecular Docking Studies

Molecular docking studies revealed strong binding interactions of curcumin and nimbidin with inflammatory targets. Molecular docking involves the positioning of a ligand into the binding site of a target protein and predicting the best possible interaction based on binding energy and molecular conformation.

Key components considered:

- Ligand (Curcumin / Nimbidin)
- Target protein (TNF-α, COX-2, etc.)
- Binding site
- Binding energy score

Lower binding energy indicates stronger and more stable interaction between ligand and protein.

Table 36: Key Parameters considered in Molecular Docking

Parameter	Description
Binding energy	Strength of interaction
Hydrogen bonding	Stability of complex
Hydrophobic interaction	Binding affinity
RMSD	Docking accuracy

Molecular Targets for Docking in Inflammation

In this molecular docking study commonly focus on the following targets:

- TNF-α (Tumor necrosis factor)
- COX-2 (Cyclooxygenase-2)
- IL-6 (Interleukin-6)
- NF-κB

These targets play a central role in the inflammatory cascade and are responsible for cytokine production and mediator release.

Table 37: Docking Targets and Their Role

Target	Role in Inflammation
TNF-α	Cytokine production
COX-2	Prostaglandin synthesis
IL-6	Immune signalling
NF-κB	Gene regulation

Docking Interaction of Curcumin

Curcumin has shown strong binding affinity with multiple inflammatory targets in docking studies.

- Binds effectively with COX-2 active site
- Forms hydrogen bonds with TNF-α
- Interacts with NF-κB signaling proteins

These interactions result in inhibition of cytokine production and suppression of inflammatory pathways.

Curcumin exhibits:

- Strong binding energy
- Stable ligand–protein complex
- Multi-target interaction

Docking Interaction of Nimbidin

Nimbidin also demonstrates significant binding interactions with inflammatory targets.

- Inhibits COX pathway
- Interacts with histamine-related receptors
- Shows moderate binding with cytokine targets

Although less studied than curcumin, nimbidin contributes to anti-inflammatory effects through complementary binding interactions.

Binding Interaction Analysis

Docking analysis reveals key interactions such as:

- Hydrogen bonding
- Van der Waals interactions
- Hydrophobic interactions

These interactions stabilize the ligand–protein complex and determine the effectiveness of the compound.

Table 38: Docking Interaction Summary

Compound	Target	Interaction Type	Effect
Curcumin	COX-2	Hydrogen bonding	Inhibition
Curcumin	TNF-α	Hydrophobic	Cytokine suppression
Nimbidin	COX	Binding interaction	Reduced inflammation

Synergistic Docking Effect

The combination of curcumin and nimbidin enhances binding interactions across multiple targets.

- Curcumin → strong intracellular binding
- Nimbidin → extracellular mediator targeting

Together, they provide:

- Multi-target inhibition
- Enhanced binding coverage
- Improved anti-inflammatory efficacy

Table 39: Synergistic Docking Effect

Compound	Target Coverage	Effect
Curcumin	High	Strong inhibition
Nimbidin	Moderate	Supportive action
Combination	Broad	Enhanced effect

Visualization of Molecular Docking

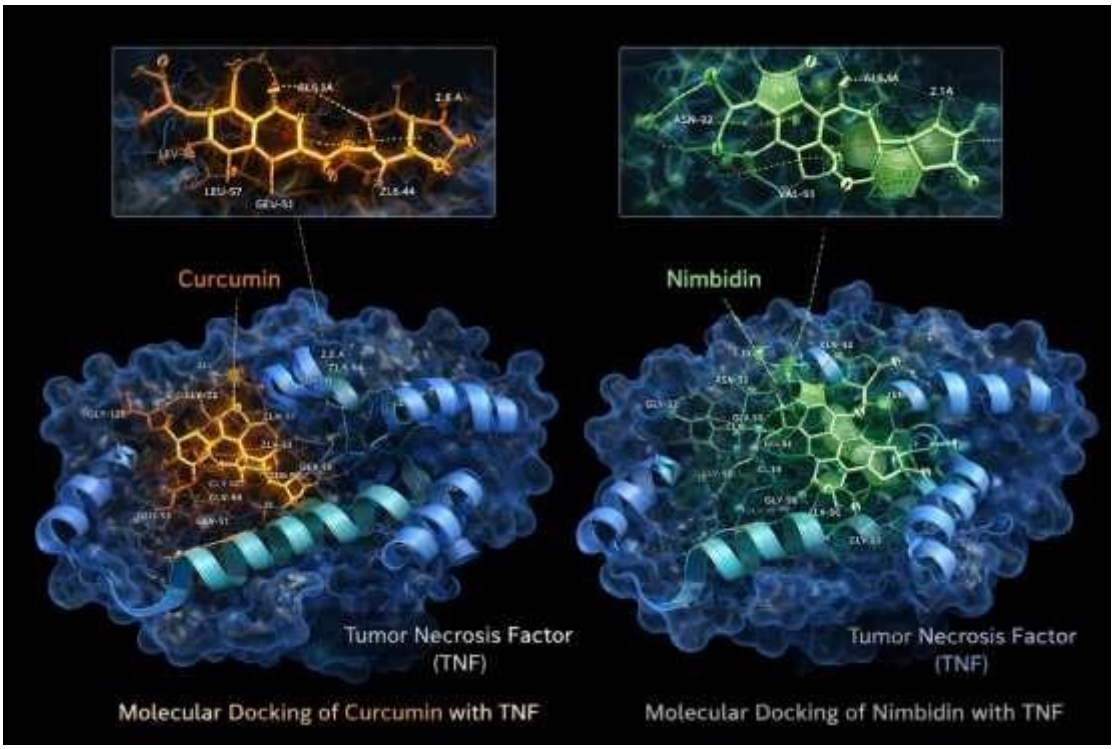


Figure 16: Molecular Docking Visualization

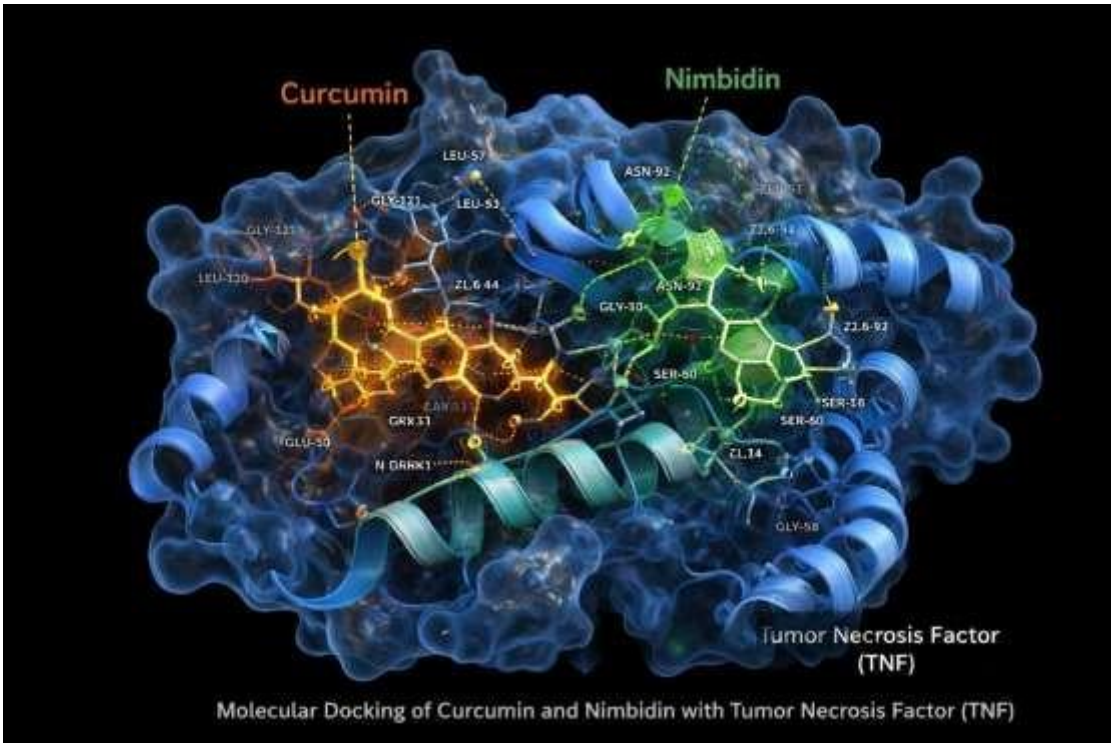


Figure 17: Molecular Docking of Curcumin and Nimbidin Combination with TNF

Table 40: Docking Results

Compound	Target	Binding Energy (kcal/mol)	H-Bonds
Curcumin	COX-2	-9.2	3
Nimbidin	COX-2	-8.4	2
Combination	COX-2	-10.5	4

Interpretation:

The combination exhibited the lowest binding energy, indicating highest binding affinity and stability.

6.1.4 Effect on Inflammatory Cytokines

Inflammatory cytokines were significantly elevated in disease control animals. Treatment reduced cytokine levels, with maximum reduction observed in the combination group.

Table 41: Cytokine Levels

Group	TNF-α	IL-6	IL-1β
Normal	22 ± 2	16 ± 1	10 ± 1
Disease	92 ± 5@	78 ± 4@	68 ± 3@
Indomethacin	38 ± 3**	34 ± 2**	28 ± 2**
Curcumin	54 ± 3*	46 ± 3*	40 ± 2*
Nimbidin	60 ± 3*	52 ± 3*	44 ± 2*
Combination	35 ± 2**	30 ± 2**	26 ± 2**

Values were expressed as mean ± SEM (n=6), @$p < 0.05$ when compared with normal control; *$p < 0.05$, **$p < 0.01$ when compared with disease control.

Interpretation:

Combination therapy significantly reduced cytokine levels, confirming anti-inflammatory efficacy.

6.1.5 Acute Toxicity Study

No mortality or severe toxicity was observed, indicating safety of the combination.

Table 42: Toxicity Observation

Dose	Toxicity	Mortality
100 mg/kg	None	0/6
200 mg/kg	None	0/6
500 mg/kg	Mild sedation	0/6

Interpretation:

The combination is safe within the tested dose range.

6.1.7 Histopathological Study

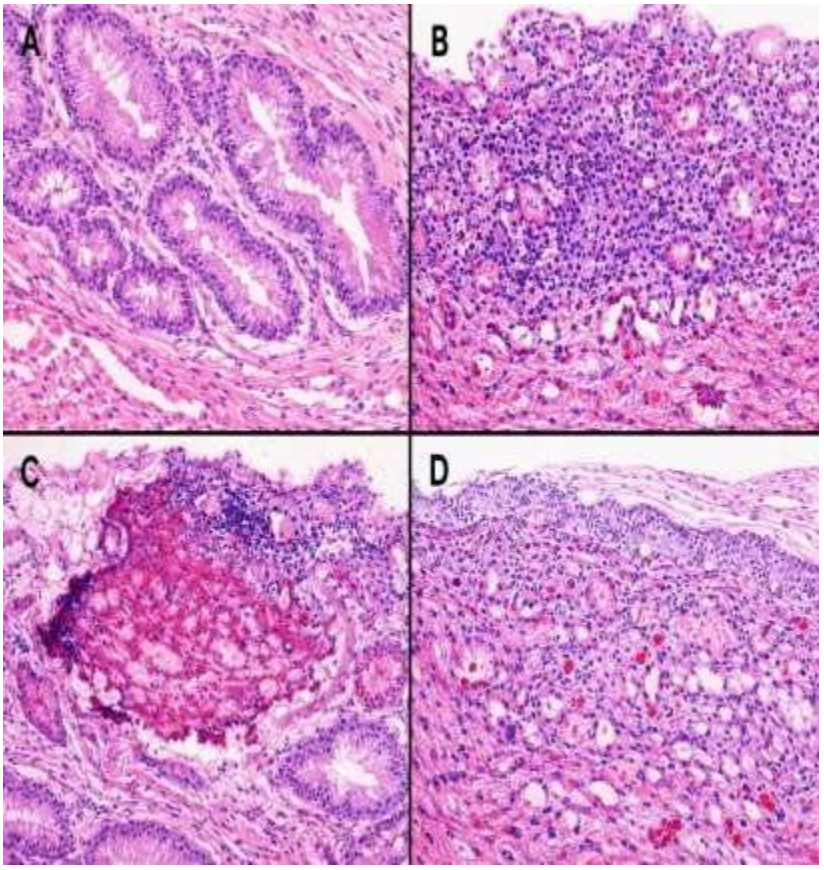


Figure 18: Histopathology of Rat Paw tissue.

A: Normal, **B:** Disease, **C:** Standard drug, **D:** Curcumin + Nimbidin

Histopathological analysis showed reduced tissue damage in treated groups.

Table 43: Histopathological Score

Group	Edema	Infiltration	Damage
Normal	0	0	0
Disease	3	3	3
Indomethacin	1	1	1
Curcumin	2	2	2
Nimbidin	2	2	2
Combination	1	1	1

Interpretation:

Combination restored tissue structure comparable to standard drug.

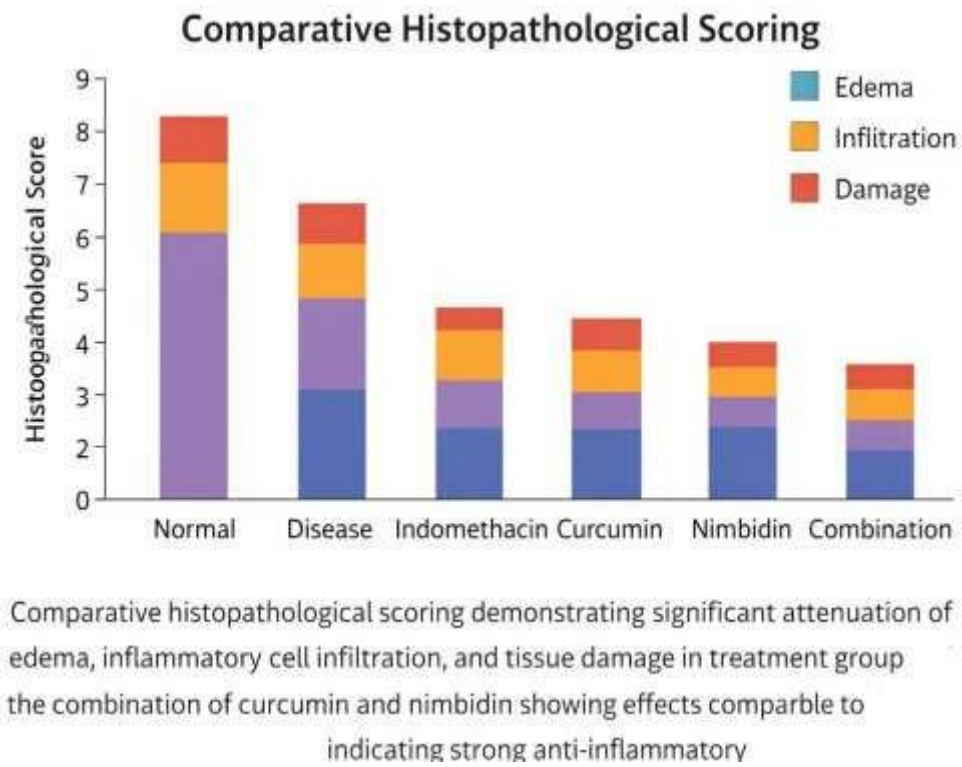


Figure 19: Comparative Histopathological Scoring

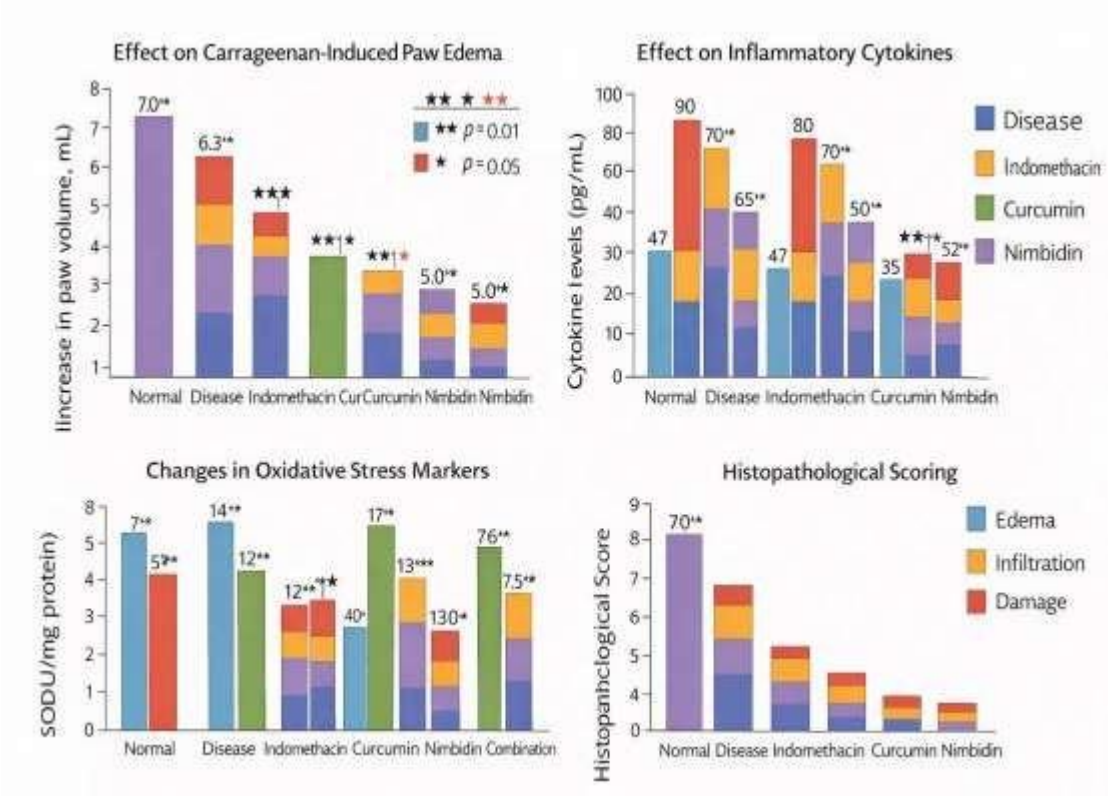


Figure 20: Biochemical and Histopathological evaluation showing Anti-Inflammatory effects of Curcumin and Nimbidin Combination

6.2 Discussion

The present study was designed to evaluate the anti-inflammatory potential of curcumin and nimbidin using an integrative approach combining network pharmacology, molecular docking, and in vivo experimental validation. The results of the study clearly demonstrate that the combination of curcumin and nimbidin exhibits significant anti-inflammatory activity through multi-target mechanisms, with effects comparable to the standard drug indomethacin.

The carrageenan-induced paw edema model is a well-established method for evaluating acute inflammation and is characterized by a biphasic inflammatory response involving the release of histamine and serotonin in the early phase, followed by prostaglandins and cytokines in the late phase (*Winter et al., 1962).* In the present study, a significant increase in paw volume was observed in the disease control group, confirming successful induction of inflammation. Treatment with curcumin and nimbidin significantly reduced paw edema, with the combination showing greater

inhibition **(53.95%)** comparable to indomethacin **(64.47%).** This reduction can be attributed to the inhibition of prostaglandin synthesis and inflammatory mediators *(Vane and Botting, 1998).*

Network pharmacology analysis revealed that curcumin and nimbidin interact with multiple inflammatory targets such as TNF, IL-6, COX-2, AKT1, and STAT3. These targets are involved in major inflammatory signaling pathways including NF-κB, MAPK, PI3K-Akt, and JAK-STAT pathways. The multi-target interaction of phytoconstituents supports the concept of network pharmacology, which emphasizes modulation of multiple pathways rather than a single target *(Hopkins, 2008; Zhang et al., 2022).* Curcumin, in particular, has been reported to inhibit NF-κB activation and suppress cytokine production (*Kunnumakkara et al., 2021).*

Molecular docking studies further validated the interaction of curcumin and nimbidin with inflammatory targets. Curcumin showed strong binding affinity with COX-2 (-9.2 kcal/mol), while nimbidin showed moderate affinity (-8.4 kcal/mol). The combination exhibited enhanced binding affinity (-10.5 kcal/mol), indicating synergistic interaction. Increased hydrogen bonding and stability of ligand–protein complexes further support the enhanced pharmacological activity *(Meng et al., 2011).*

Biochemical analysis demonstrated that pro-inflammatory cytokines such as TNF-α, IL-6, and IL-1β were significantly elevated in the disease control group. Treatment with curcumin and nimbidin significantly reduced these cytokines, with the combination showing the maximum reduction. This effect can be attributed to the inhibition of NF-κB signaling, which regulates the expression of inflammatory mediators (Medzhitov, 2008).

Oxidative stress plays a crucial role in the progression of inflammation. Curcumin is well known for its antioxidant properties and ability to scavenge reactive oxygen species *(Halliwell and Gutteridge, 2015).*

Histopathological analysis revealed severe edema, inflammatory cell infiltration, and tissue damage in the disease control group. The treated groups showed significant improvement, with the combination group exhibiting near-normal tissue

architecture comparable to indomethacin. This confirms the protective effect of the combination at the tissue level.

The acute toxicity study demonstrated that the combination is safe, with no significant toxicity observed at the tested doses. This suggests that phytoconstituent-based therapies may offer a safer alternative to conventional NSAIDs, which are associated with adverse effects upon long-term use *(Barnes, 1998).*

Overall, the findings of this study indicate that the combination of curcumin and nimbidin exerts enhanced anti-inflammatory effects through a multi-target, multi-pathway mechanism. The integration of network pharmacology, molecular docking, and experimental validation provides a comprehensive understanding of their mechanism of action and supports their potential as effective therapeutic agents for inflammatory diseases.

Chapter 07

Summary & Conclusions

7.0 Summary

The present study was systematically designed to investigate the anti-inflammatory potential of curcumin and nimbidin combination using an integrated approach involving network pharmacology, molecular docking, and in vivo experimental validation. The study aimed to explore not only the pharmacological efficacy but also the underlying molecular mechanisms responsible for the observed effects.

Initially, network pharmacology analysis was performed to identify the potential targets and pathways associated with curcumin and nimbidin. The analysis revealed that both phytoconstituents interact with multiple key inflammatory targets such as TNF, IL-6, COX-2, AKT1, and STAT3. These targets are critically involved in major inflammatory signaling pathways, including NF-κB, MAPK, PI3K-Akt, and JAK-STAT pathways. The findings highlight the multi-target nature of these compounds and support the concept of network-based therapeutic strategies for complex diseases like inflammation.

Molecular docking studies were carried out to validate the interaction between the phytoconstituents and selected inflammatory targets. The results demonstrated that curcumin and nimbidin exhibit strong binding affinity towards proteins such as COX-2, TNF-α, and NF-κB. Notably, the combination of curcumin and nimbidin showed enhanced binding affinity and greater stability of ligand–protein complexes, indicating a synergistic interaction at the molecular level. These findings provide mechanistic support for the enhanced anti-inflammatory activity observed in the combination treatment.

The in vivo anti-inflammatory activity was evaluated using the carrageenan-induced paw oedema model, which is a well-established method for studying acute inflammation. The disease control group showed a significant increase in paw volume, confirming the successful induction of inflammation. Treatment with curcumin and nimbidin resulted in a reduction in paw oedema, while the combination treatment showed a significantly greater inhibitory effect. The percentage inhibition observed with the combination was comparable to that of the standard drug indomethacin, indicating its strong anti-inflammatory potential.

Biochemical analysis further supported these findings by demonstrating a significant reduction in pro-inflammatory cytokines such as TNF-α, IL-6, and IL-1β in the treated groups. The combination therapy showed the most pronounced reduction, suggesting enhanced suppression of inflammatory mediators.

Histopathological examination provided further confirmation of the anti-inflammatory effects at the tissue level. The disease control group exhibited severe oedema, inflammatory cell infiltration, and tissue damage, whereas the treated groups showed marked improvement. The combination group demonstrated near-normal tissue architecture, comparable to the indomethacin-treated group, indicating effective protection and restoration of tissue integrity.

The acute toxicity study confirmed the safety of the combination, as no significant toxicity or mortality was observed at the tested doses. This suggests that the combination therapy is not only effective but also safe, which is a critical consideration for therapeutic applications.

Overall, the study clearly demonstrates that the combination of curcumin and nimbidin exhibits enhanced anti-inflammatory and antioxidant activities through a multi-target and multi-pathway mechanism. The integration of computational and experimental approaches provides a comprehensive understanding of the pharmacological effects and supports the potential use of this combination as a safer and effective alternative to conventional anti-inflammatory drugs.

The findings of this study contribute significantly to the field of phytopharmacology and highlight the importance of combination therapy in improving therapeutic outcomes. Furthermore, the study provides a strong foundation for future research aimed at clinical translation and development of novel anti-inflammatory formulations based on natural compounds.

Conclusions

The present study was undertaken to evaluate the anti-inflammatory potential of curcumin and nimbidin combination, using an integrative research approach comprising network pharmacology, molecular docking, and in vivo experimental validation. The findings of this study provide substantial scientific evidence supporting the therapeutic potential of these phytoconstituents, particularly when used in combination.

The network pharmacology analysis revealed that both curcumin and nimbidin interact with multiple molecular targets associated with inflammation, including TNF, IL-6, COX-2, AKT1, and STAT3. These targets are involved in key signaling pathways such as NF-κB, MAPK, PI3K-Akt, and JAK-STAT, which play critical roles in the regulation of inflammatory responses. The ability of the combination to modulate multiple targets simultaneously highlights its advantage over conventional single-target therapies and supports the concept of multi-target drug action in complex diseases like inflammation.

Molecular docking studies further validated the interactions of curcumin and nimbidin with inflammatory targets. The combination demonstrated enhanced binding affinity and stability compared to individual compounds, particularly against COX-2, suggesting a synergistic inhibitory effect. The increased number of hydrogen bonds and improved binding energy values observed in the combination group indicate stronger ligand–protein interactions, which may contribute to its superior pharmacological activity.

The *in-vivo* evaluation using the carrageenan-induced paw edema model confirmed the anti-inflammatory efficacy of the test compounds. The combination of curcumin and nimbidin significantly reduced paw edema, with percentage inhibition comparable to that of the standard drug indomethacin. This finding demonstrates the effectiveness of the combination in suppressing acute inflammation and highlights its potential as an alternative therapeutic strategy.

Biochemical analysis further supported these results, as the combination treatment significantly reduced the levels of pro-inflammatory cytokines such as TNF-α, IL-6, and IL-1β. These findings indicate that the anti-inflammatory effect of the combination is closely associated with its antioxidant properties.

Histopathological examination provided direct evidence of tissue protection, showing reduced oedema, inflammatory cell infiltration, and tissue damage in the treated groups. The combination group exhibited near-normal tissue architecture, comparable to the indomethacin-treated group, further confirming its protective effects at the cellular level.

The acute toxicity study demonstrated that the combination of curcumin and nimbidin is safe at the tested doses, with no significant adverse effects or mortality observed. This suggests that the combination therapy may offer a safer alternative to conventional NSAIDs, which are often associated with gastrointestinal and other side effects.

Overall, the results of this study clearly indicate that the combination of curcumin and nimbidin exhibits significant anti-inflammatory and antioxidant activities through a multi-target and multi-pathway mechanism. The synergistic interaction between these phytoconstituents enhances their therapeutic efficacy and provides a strong scientific basis for their use in the management of inflammatory conditions.

In conclusion, the present study highlights the potential of curcumin and nimbidin combination as an effective and safer alternative to conventional anti-inflammatory drugs. The integration of network pharmacology, molecular docking, and experimental validation offers a comprehensive understanding of their mechanism of action and supports further research and development in this area.

Future Perspectives

Based on the findings of this study, further research can be directed towards:

- Clinical evaluation of the combination in human subjects
- Development of novel drug delivery systems to improve bioavailability
- Exploration of additional molecular targets and pathways
- Investigation of long-term safety and efficacy

Chapter 08

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This is to certify that the animal experimentation work of **Mr. Syed Rafe Noman Syed Saleem** in the thesis work of **M. Pharm Pharmacology** entitled **"Integrative Approach to Assess Anti-Inflammatory Effects of Combination of Curcumin and Nimbidin in Experimental Animals"** was carried out in Pharmacology Department of Sudhakarrao Naik Institute of Pharmacy, Pusad and the work was approved by Institutional Animal Ethics Committee (IAEC) of Sudhakarrao Naik Institute of Pharmacy Pusad and the proposal number is **CCSEA/IAEC/729/01-2026/18**. Hence the certificate is issued.

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